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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter 2: Not started. Will write last/almost last
- Chapter ??: Some ideas for content, but no words written yet.
- Chapter ??: Approaching first draft. Lots of work is still required to make clear that terms are properly defined.
- Chapter 4: Passed first drafting. Awaiting comments.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's comments.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits.
- Chapter ??: Not written or properly planned
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- Chapter ?? : Not written or properly planned. Might not make it past planning stage
- Chapter ?? : Not written or properly planned. May be relegated to the appendix
- Chapter ?? : Significant amounts of thought has already gone into this.
- Chapter ?? : Not started. Will write last/almost last
- Appendix ?? : Passed first drafting.
- Appendix ?? : Mostly written.
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Declaration

Acknowledgements

Chapter 1

Abstract

Chapter 2

Introduction

This chapter should clearly motivate the need for gravitational wave measurements in the context of cosmology and astrophysics. It will (briefly) explain the types of sources that the detectors will be measuring (including a bit about LISA). It will talk about tests of general relativity, measurements of the age of the universe, multi-messenger measurements of the neutron star equation of state (its a combination of basically every field of physics), and so on... (I will read up more specific examples, but having large numbers of observations will tell us something about astrophysics). Although I'm not sure if I will mention dark matter in great depth, it probably deserves a mention since its one of the great mysteries of physics and gravitational wave detections may give insights into it. This will be an expanded version of the introduction to most papers written about gravitational wave detections, which hopefully will give you an idea of the tone of this chapter. It will be a relatively short chapter (about 5-10 pages?), but its vital in giving the rest of the work a reason to exist (some people do study graviational wave detectors as tools for probing the quantum nature of light, but most people will not see this as a good enough

reason to spend billions of dollars developing them, heck not even everyone in the LVK collaboration thinks this is a good motivation for building the instruments!). Finally it will end with an explanation of each chapter in the thesis. This chapter will not cover the working principles of an interferometer

Chapter 3

Gravitational Wave Interferometers

Chapter 4

A Prototype High Power Laser for LIGO's Fourth Observation Run

4.1 Introduction

The LIGO detectors require high input power, nominally 125 W at 1064 nm [1], to decrease the shot noise which limits its sensitivity at frequencies above 100 Hz. The current laser cannot reach the required power without generating excess noise. In this chapter, the new laser layout is discussed. The PMC throughput power, beam quality, intensity noise, pointing noise and frequency noise of the prototype laser were measured. It was found that this layout provided a suitably low noise laser. This laser was able to produce 100 W of intensity stabilised light. While the 125 W nominal power level was not obtained, this layout provides more power than the current laser. With additional work in optimising the layout, the maximum output power of this laser may be increased.

If a laser is in a coherent state, the shot noise associated with detecting the laser is $\sigma = \sqrt{\frac{hc}{\lambda}P}$, so the signal to noise ratio at high frequencies scales with $\sqrt{P}$. Alternatively, to decrease the shot noise limit the wavelength could be decreased; however, the LIGO detectors operate at 1064 nm due to the availability of mirror coatings with low absorption and low thermal noise. Future interferometers may operate at longer wavelengths due to the potential reduction of mirror coating thermal noise that can be achieved at longer wavelengths [2].

In the field of gravitational wave detection, high power refers to lasers with powers of the order 100 W. High power is a loosely defined term as lasers ranging from 100 W to 100 kW are defined as high power in the literature [3–13]. As the power is increased, other properties of the laser tend to be degraded. It needs to operate uninterrupted on year-long timescales; this poses a major engineering challenge.

The laser for LIGO must be continuous wave, linearly polarised, have high spatial purity, low intensity noise and low frequency noise. These requirements are detailed in [1]. To overcome the technical challenges that come with the high power and low noise requirements for the LIGO laser, the master-oscillator-power-amplifier (MOPA) topology was selected. This type of laser is based on a high performance seed laser which is then amplified. As amplifiers are linear and coherent, the low noise properties of the seed laser characterises the noise of the high power beam.

LIGO’s original laser used a high power oscillator (HPO) as the means for obtaining the target power. The HPO was a ring cavity containing laser crystals that was seeded by a 35 W low noise laser. This design went through extensive testing and refinement at the AEI [14] and was installed as part of LIGO’s pre-stabilised-laser

(PSL). However, due to jitter noise from the water cooling of the crystals within the HPO coupling to the differential arm motion signal (DARM), as well as one HPO being destroyed by accident, the HPO was bypassed and the LIGO detectors operated using the 35 W laser [15].

To reach higher powers during LIGO's third observation run, instead of the HPO, a neoVAN-4S single-pass amplifier was used. This stepped up the laser power from 35 W to 70 W [16, 17]. The next stage of development was a complete redesign of the LIGO laser based on using two upgraded versions of the neoVAN-4S. The neoVAN-4S-HP amplifier showed promising results [12, 13], so a prototype laser was constructed and tested at LIGO Livingston. This design will be installed as the new PSL for the start of LIGO's fourth observation run [15].

4.2 Concept for LIGO's New Laser

LIGO will use a MOPA laser that is seeded by a non-planar ring oscillator (NPRO). The NPRO beam will be amplified by two single-pass amplifiers in series and be followed by a mode cleaner cavity to produce a low-noise high-power laser in the fundamental mode. This concept is shown in Figure 4.1. As the NPRO has good free running noise characteristics the amplified beam will as well. Results from a setup using a single neoVAN-4S-HP showed these amplifiers can be continuously operated for many tens of days [13]. This led to the proposal to construct the PSL out of two of these in series. A similar set up using a 35 W seed laser and two neoVAN-4S-HPs was able to generate 195 W of light [12] for nearly 80 days with no significant drops in power.

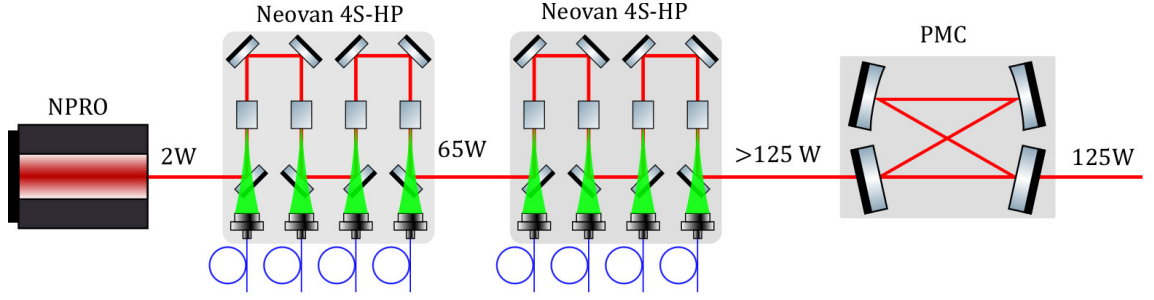


Figure 4.1: Conceptually, the PSL will be made up of four components: the NRPO, two neoVAN-4S-HP amplifiers and the PMC cavity. The NPRO provides seed light with good noise characteristics. These noise characteristics are preserved by the amplifiers, so the resulting beam will have high power and low noise. The PMC filters junk light and RF intensity noise so a spatially pure mode can be sent to the interferometer.

4.2.1 Non-Planar Ring Oscillators

The first element of the PSL is a non-planar ring oscillator [18–20]. The NPRO provides a low noise laser making it a suitable seed for a MOPA system. This is a single frequency laser constructed from one Nd:YAG crystal which acts as both the gain medium and the cavity, thus it produces a laser with a high amount of frequency stability. The NPRO used was a Mephisto, built by Coherent, and it can produce to 2W of 1064 nm light.

4.2.2 The neoVAN-4S-HP Solid-State Amplifier

To amplify the NPRO beam, two neoVAN-4S-HP amplifiers will be used. The neoVAN-4S-HP is a single-pass, solid-state amplifier manufactured by Neolase. It is constructed from four Nd : YVO₄ crystals which are pumped by fibre coupled,

low noise laser diodes. It should be seeded by s-polarised light at 1064 nm. These gain mediums are in contact with a metal plate which is cooled. This keeps jitter noise low as flowing water is not directly in contact with the gain medium. As these amplifiers have low noise pump-diodes and will be seeded by an NPRO, the noise of the light produced by them should be determined by the NPRO.

4.2.3 The Pre Mode Cleaner Cavity

The pre mode cleaner (PMC), see Figure 4.2, is used as the hand-off point between the PSL and the rest of the interferometer. Light transmitted by this cavity is used as the input beam for the interferometer. The PMC is a bow tie cavity with an FSR of 9 MHz, thus it passively filters RF intensity noise. Control sidebands are provided after the PMC but before the IMC, meaning that the IMC cannot be used to provide this stabilisation. The PMC is locked to the fundamental mode so the beam that is delivered to the rest of the interferometer will have a low amount of higher order mode content. This is important as the amplifiers produce light with spatial defects. The PMC has been redesigned [21] to prevent contamination due to glue on its mirrors and to make maintaining the PMC easier. The PMC design no longer includes a vacuum enclosure.

4.3 Layout of the Prototype Laser System

An optical layout based on the concept described in Section 4.2 was constructed. A sketch of the layout is shown in Figure 4.2. In addition to the main beam path, there are low power diagnostic beams after each amplifier that can be used to characterise the laser. The prototype layout did not include the tabletop frequency stabilisation

servo (TTFSS) so this has not been included in the layout description. However, there is space for the TTFSS on the optical table.

The NPRO beam passes through some optics to clean up the polarisation. Transmission through a steering mirror is used to monitor the power of the NPRO. An electro-optic modulator (EOM) is used to produce 35.5 MHz sidebands for the Pound-Drever-Hall (PDH) scheme used to lock the PMC. A Faraday isolator ensures light reflected back from the amplifiers does not damage the NPRO. The beam passes through a lens to mode match the beam to the first amplifier. Two mirrors are used to align the NPRO beam to the amplifier. A half-wave plate ensures the polarisation into the first amplifier is set correctly. These optics are shown in Figure 4.3.

After the first amplifier, the beam passes through a high power attenuation stage, which consists of a half wave-plate and two thin film polarisers (TFP), to allow for safe set up and low power operation. When setting up this layout, the amplifiers must be run at maximum power since they exhibit significant thermal lensing. The high power attenuators are not envisioned being used during observation runs. The laser is then directed through a Faraday isolator and wave-plate before reaching the second amplifier. The mode matching lenses for the second amplifier are on either side of the Faraday isolator. Two mirrors immediately before the second amplifier are used for its alignment. The transmission through one of the steering mirrors before the second amplifier is used for monitoring the beam produced by the first amplifier. This section of the layout is shown in Figure 4.3.

After the second amplifier, the beam goes through a high power attenuation stage before passing through an acousto-optic modulator (AOM). This AOM is used for

controlling the power of the laser. Downstream of the AOM, a pick-off beam is used to measure the properties of the laser produced by the second amplifier. Before entering the PMC, the beam is transmitted through a TFP so that the beam is set to be p-polarised. This part of the layout can be seen in Figure 4.4.

The beam produced by the second amplifier is then coupled to the PMC; this requires two mode matching lenses and two steering mirrors. There are three beams transmitted by the PMC: a high power one, which would be sent to the interferometer, and two low power ones for stabilising the frequency and intensity of the laser. The low power beams contain of the order 100 mW of light. The intensity stabilisation servo (ISS) inner-loop photodiodes are in a light-tight enclosure. One of the photodiodes is used as the in-loop sensor while the other is used as the out-of-loop sensor. The PMC is shown in Figure 4.5. The reflected light is used to lock the PMC; this path can be seen in Figure 4.4.

The diagnostic bread board (DBB) [22,23] is integrated into the layout via several pick-off points and can be used to monitor the status of both amplifiers. The DBB enables measurements of intensity noise, frequency noise, beam pointing noise and higher order mode content. The DBB is shown in Figure 4.6. The DBB consists of a cavity, motorised lenses, an automatic alignment system and several photodiodes. Despite having motorised lens mounts in the DBB, static mode matching lenses are required in the diagnostic beam paths. To characterise the NPRO beam, mirrors on magnetic mounts can be inserted into the layout.

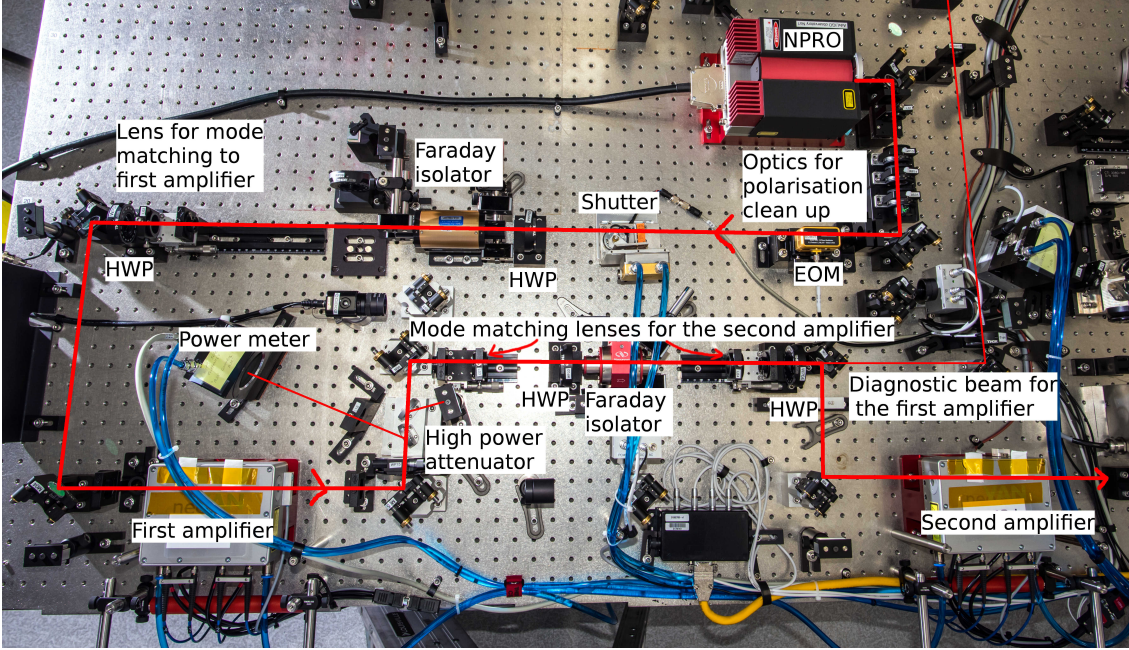


Figure 4.3: Photograph of the PSL prototype showing the beam path (red lines) from the NPRO (top right) to the second amplifier (bottom right). The beam path up to the first amplifier includes optics for setting the polarisation of the beam, an EOM, a Faraday isolator and lenses for mode matching. After the first amplifier, the beam passes through a high power attenuator, mode matching lenses and a Faraday isolator. The first amplifier (bottom left) is monitored using a low power pick-off beam (centre right).

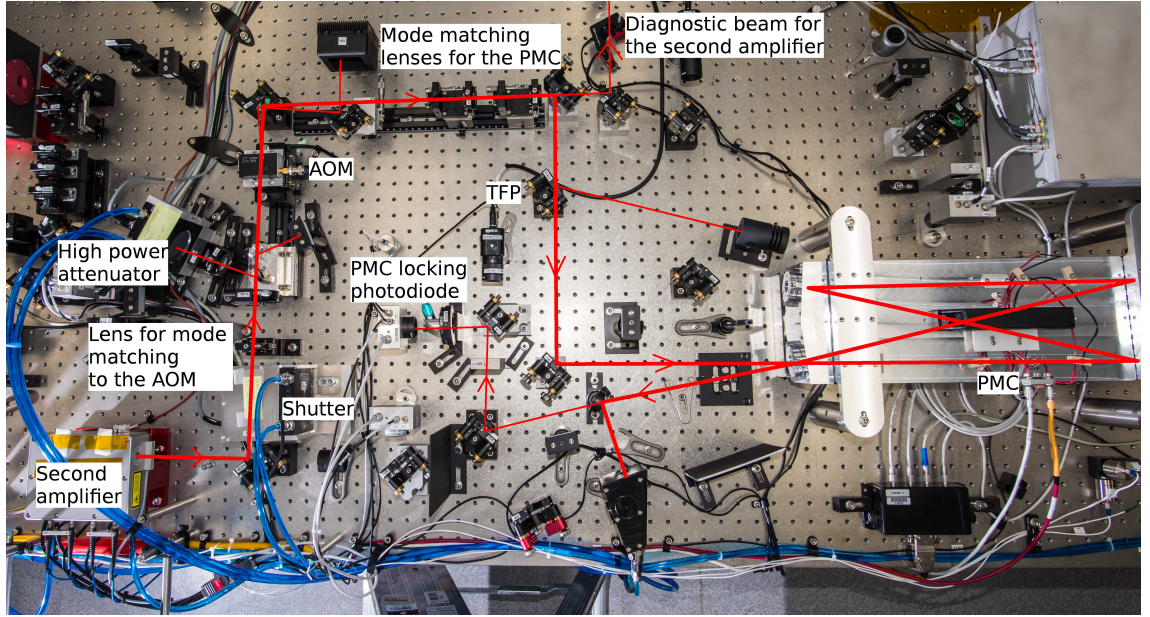


Figure 4.4: Photograph of the layout showing the beam path (red lines) from the second amplifier (bottom left) to the input of the PMC (centre right). This beam path includes a high power attenuator, an AOM, mode matching lenses for the PMC and a TFP. The PMC can be seen more clearly in Figure 4.5. The path taken by light reflected from the PMC is shown. This light is used to lock the PMC. A low power beam for monitoring the second amplifier is obtained using the transmission through a steering mirror (centre top) for the alignment to the PMC.

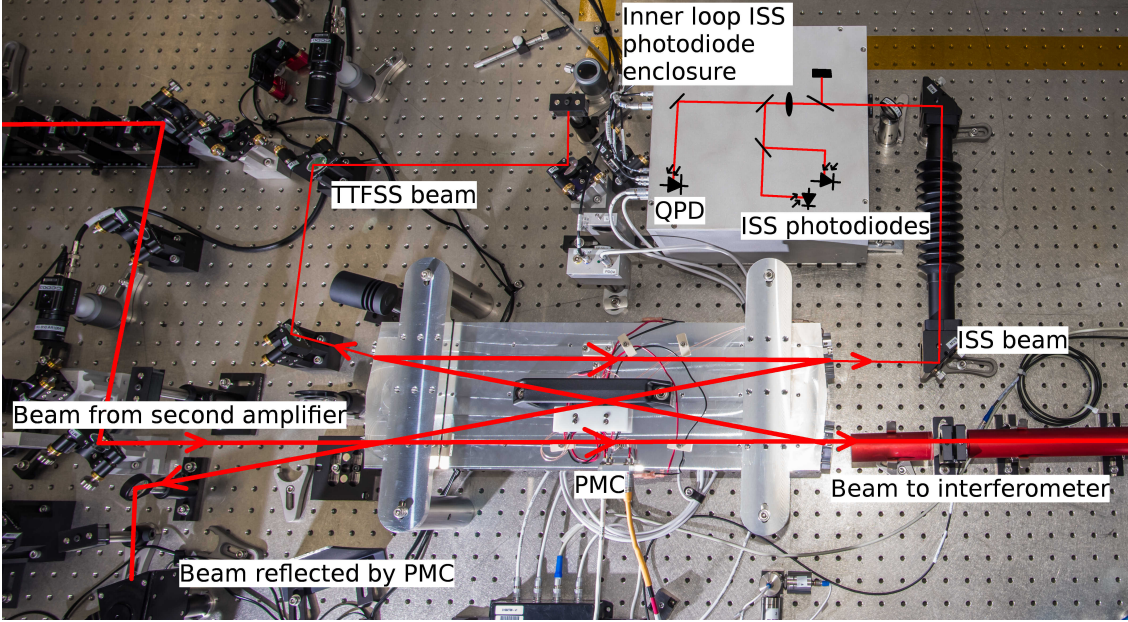


Figure 4.5: Photograph of the PMC (centre) and the inner ISS photodiode box (top right). The beam (shown in red) from the second amplifier enters the PMC. The majority of this light is sent to the interferometer. A small fraction of the light in transmission of the PMC is used to stabilise the intensity and frequency of the laser. These beams are labelled as the ISS beam and the TTFSS beam. The ISS beam is directed into a light tight enclosure. This enclosure contains two photodiodes which are for measuring the intensity noise of the laser. One of these is an in-loop sensor; the other is an out-of-loop sensor. A QPD can be used to monitor the alignment of the ISS photodiodes. The reflected beam from the PMC is made up of the junk light in the beam from the second amplifier.

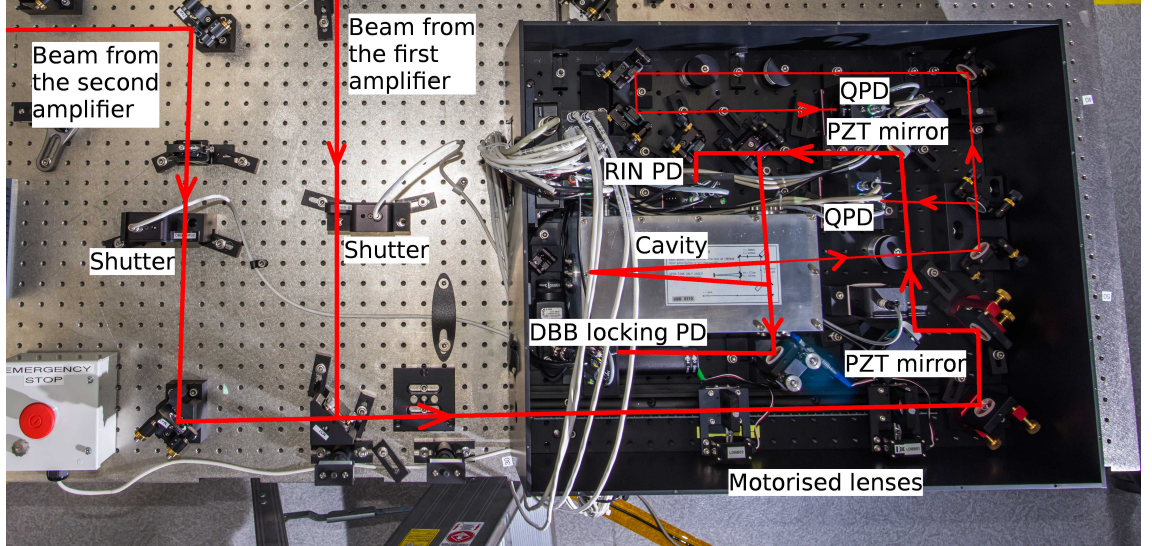


Figure 4.6: Photograph of the DBB with the beam path highlighted in red. Low power beams are used to measure properties of the laser produced by both amplifiers. The shutters control which amplifier will be characterised. The DBB contains motorised lenses and PZT mirrors so that remote mode matching and automatic alignment can be performed. The intensity noise of the beam can be measured with the RIN photodiode. The DBB locking photodiode is used to lock the cavity to the beam. This photodiode is also used during mode scans to measure the higher order content of the beam. QPDs separated by 90° of Gouy phase are used to measure the pointing noise of the incoming beam.

4.4 The Power Transmitted by the PMC

The PSL should deliver 125 W to the interferometer [1]. The power transmitted through the PMC is determined by the power in the beam generated by the amplifiers and the amplified beam's distortion. The beam's distortion is affected by the input alignment, mode matching and thermal lensing. Increasing the pump-diode current increases the power that is produced by the amplifier as well as the amount of thermal lensing. The flow rate of coolant can be increased to mitigate thermal lensing; however, this can generate jitter.

The prototype layout was able to deliver a maximum of 111 W through the PMC during a 55 hour trial run. This is shown in Figure 4.7. The power was measured with recently calibrated power meters. The power emerging from the PMC dropped to 107 W during the course of the trial run, while the power incident onto it remained ~ 125 W. Changes in the lab temperature and humidity caused slight alignment changes. This resulted in the power decreasing. As the laser will be in a controlled environment, this effect will be less severe when it is installed.

To increase the output power, the mode matching to the PMC could be improved and the amplified beam's distortion could be reduced. Mode matching between the final amplifier and the PMC could be improved at the percent level. Increasing the flow rate would reduce the beam's distortion and may allow for higher pump-diode currents, however concerns of jitter noise meant that the flow rate was kept low. With higher flow rates, a beam with 137 W of power and a higher mode content of 4.6% was produced by the amplifiers.

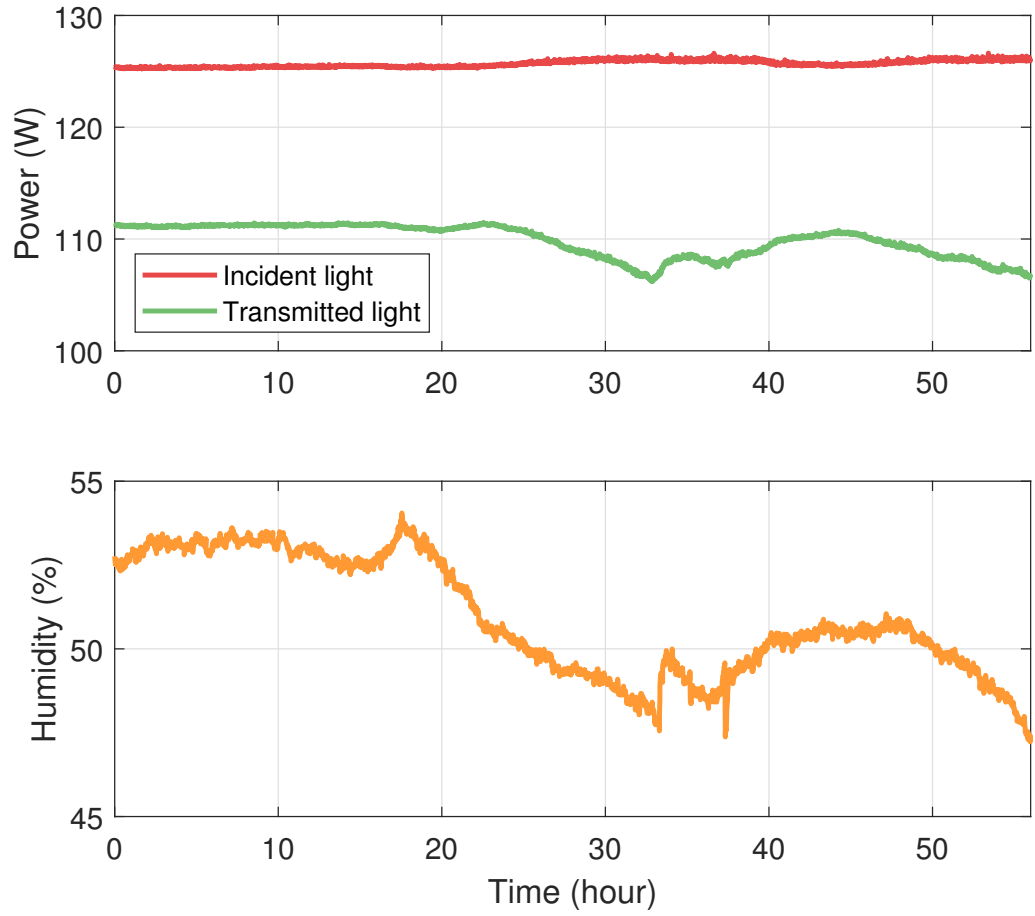


Figure 4.7: The top panel shows the power that was incident on the PMC and the power emerging from the PMC over a 55 hour trail run. The light produced by the amplifiers did not change significantly for the duration of this measurement, however the transmitted light through the PMC dropped from 111 W to, at its lowest, 107 W. This is likely due to the environmental conditions in the lab changing. The bottom panel shows the humidity of the lab. Changes in the humidity, which can be caused by a temperature change, coincide with changes in the power emerging from the PMC.

To actuate on the laser’s power, some of the incident light will need to be ‘dumped’ to provide headroom for the actuator to increase the beam’s power. This means that the full power will be reduced by ~ 10 W when the intensity stabilisation is implemented.

Long term measurements should be taken, however the results in [12] show these amplifiers can be run continuously for at least 80 days. LIGO Hanford measured the input laser power dropping from 55 W to 52 W over the course of one year (cite Nina’s new paper); this is likely due to mode mismatches as a result of changes in the set-up’s thermal state.

4.4.1 The Effect of Cooling on the Amplifier’s Power and Beam Quality

As the quality of the beam emerging from an amplifier is affected by the temperature of the amplifier’s head, an experiment was performed to determine a suitable temperature to run the amplifiers at. Due to the jitter noise witnessed at the Hanford detector, the flow rates were set as low as possible before the amount of power in the fundamental mode dropped significantly. As the Nd : YVO₄ crystals are cooled non-uniformly, the thermal lens is non-circular. For small amounts of thermal lensing, the beam will become slightly astigmatic.

The cooling layout that was used to find a suitable amplifier head temperature to operate with is shown in Figure 4.8. The diode box and amplifier head were connected in parallel, and each amplifier was on a separate cooling loop so that the amplifier head temperatures could be independently controlled. The flow rate to the

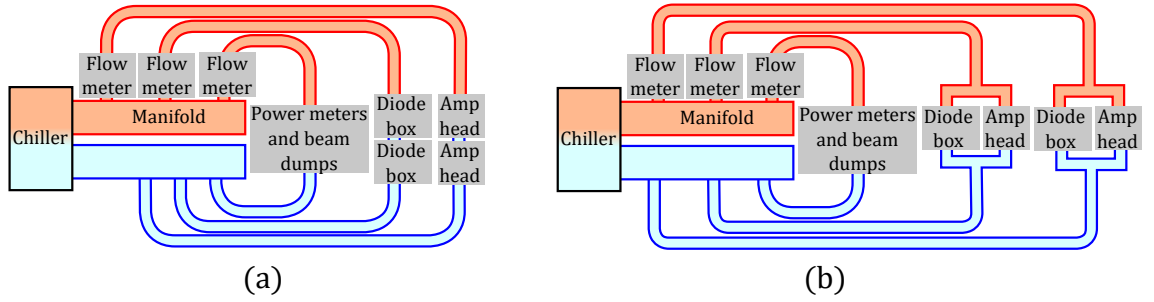


Figure 4.8: Cartoons of water cooling layouts for the prototype layout. Panel (a): The laser will be run with the amplifier heads cooled in series with each other to reduce jitter. Panel (b): To perform the amplifier head temperature experiment, the water cooling was arranged so that the amplifiers temperature could be independently controlled.

first amplifier was kept high to a high quality input beam for the second amplifier. The HOM content for the beam entering the second amplifier was 3.8%. The second amplifier and diode box flow rate was decreased and the temperature of the diode head was monitored. The higher order mode content was measured with the DBB by averaging over sixty mode scans. The power was measured with a high-power power meter.

The change in beam quality and output power as a function of amplifier temperature is shown in Figure 4.9. At 22.3°C the higher order mode content was $\sim 4.5\%$ and the amount of power in the fundamental mode was $\sim 127\text{ W}$. Figure 4.10 shows that changes in the second order mode make up the majority of the HOM content change. This is due to the beam becoming more astigmatic as the amplifier head heats up. It was decided that to keep the amount of power in the fundamental mode high, the diode heads would be operated such that their temperature did not

rise above $\sim 27^{\circ}\text{C}$.

To be able to keep the flow rate as low as possible on the table, the water cooling was set up so that the amplifiers heads were in series with each other. The diode boxes were on another loop to the heads. The cooling scheme that will be used is shown in Figure 4.8. Meters which monitor the flow rate of the coolant will be placed on the return side of the system rather than the supply side. By having the flow meters on the return side, they will measure a drop in flow if there is a leak, allowing for the water supply to be cut.

4.4.2 Amplifier Alignment Procedure

To obtain high beam power and low higher order mode content, the alignment of the input beam to the amplifier needs to be optimised. The output power of the amplifier may increase at the expense of higher order mode content, so the beam profile and the beam power must be monitored when aligning the amplifier. Optimising the alignment to the amplifier is an iterative process, so with each increase in power a small change in the amplifier's alignment is required.

Initially, the amplifier was switched off. This protects it against damage as well as preventing the possibility of stray light beams with several tens of Watts of power being in the lab. The beam path between the NPRO and the exit of the first amplifier were inspected to make sure the beam path was clear and terminated with dumps.

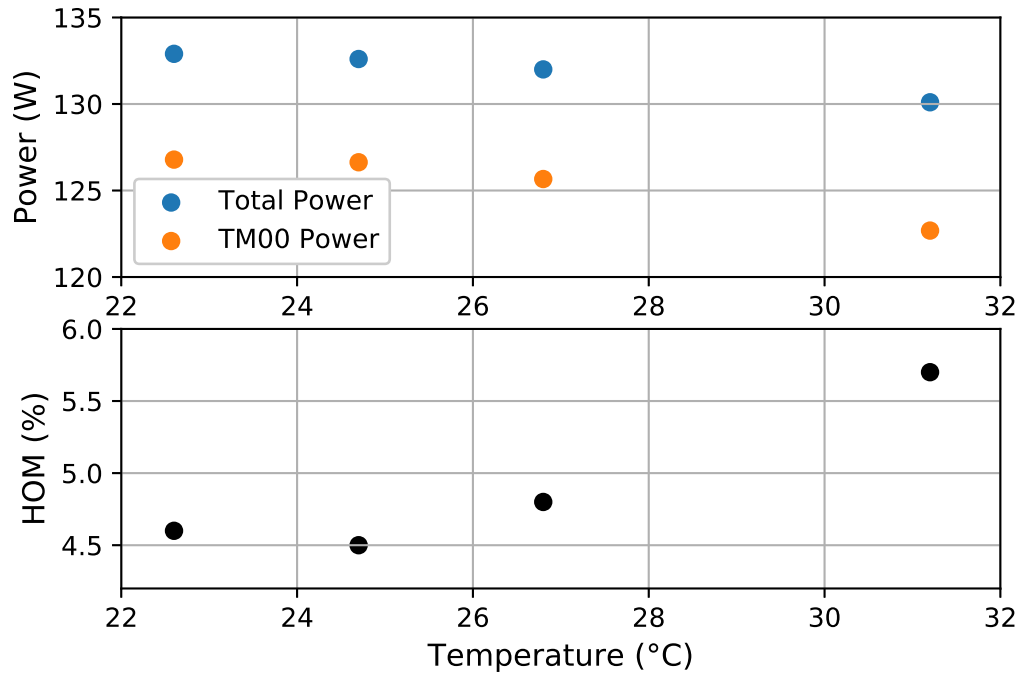


Figure 4.9: The top panel shows that as the second amplifier head's temperature increases the power produced by it decreases. The bottom panel shows that the higher order mode content increases as the temperature of the amplifier increases. This means that the power in the fundamental mode decreases as the temperature decreases due to two effects: less overall power and worse beam quality.

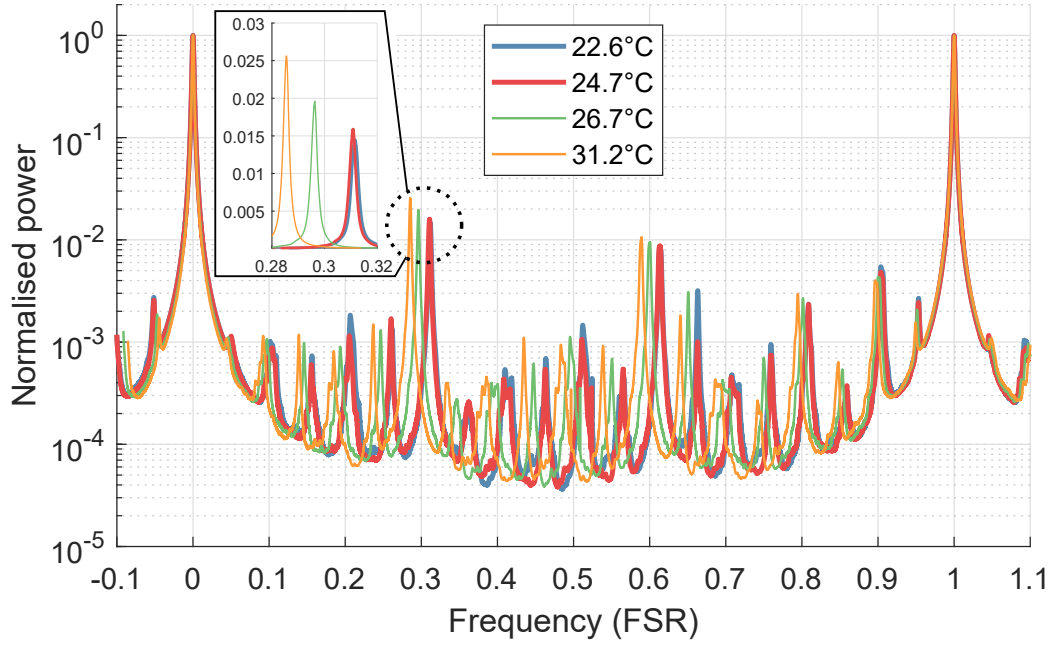


Figure 4.10: Mode scans were performed with the DBB for each amplifier head temperature. The second order mode for this cavity is located at ~ 0.3 FSR; its exact location drifts due to thermal expansion of the cavity. As the temperature of the amplifier head changes, the second order mode content of the beam changes at the percent level. This second order mode height is shown in the inset. The second order mode changes due to the beam's astigmatism changing.

Next, the NPRO was set to produce 1.3W of light. The power of the light was measured on the input side of the amplifier. The beam was roughly aligned so that the spot would hit the centre of the amplifier's input aperture using the two mirrors that immediately precede the amplifier. The power meter was placed on the exit side of the amplifier, and the input alignment was adjusted until the power emerging from the amplifier was the same as the input power. This indicates that the beam will be roughly aligned and some amplification will occur when the pump-diodes are turned on. The high power attenuator immediately after the amplifier was set to dump all the light. The initial mode matching was based on the manufactures specifications, and the input light was set to s-polarisation.

The pump-diode currents were increased until the amplifier 'switched on'. This was at 2 A of current through each of the pump-diodes. Amplification begins when there is sufficient pump light to cause enough population inversion that stimulated emission occurs. At this threshold, the output power was maximised by optimising the alignment. Then, the mode matching was adjusted in tandem with correcting for mis-alignments caused by non-perfect centring of the beam on the mode matching lens.

A fast low-power power meter was used to monitor the beam transmitted through the high power attenuator. Compared to the high-power meter, the low-power meter drifted less and responded to changes in power faster. This gave quicker feedback and allowed for easier alignment.

A Wincam was set up to monitor the beam profile. If the amplifier is significantly misaligned and mode mismatched, the beam profile will become distorted. Large

misalignment will cause the beam profile to look lopsided, and smaller misalignments will cause the beam profile to have shoulders.

Once the amplifier had been optimised for a low pump-diode current, the pump-diode current was incremented in small steps of roughly 1-2A. Each time this was done, the thermal lensing increased. Due to the non-uniform way in which the heating occurred, the input alignment was re-optimised each time as was the input mode matching. By going in small steps of diode current, the optimal alignment was safely achieved. The amplifier should be operated at high diode currents, $\sim 9\text{A}$, so at low currents it was sufficient to be within 1% of optimal.

To determine when the alignment was optimal, mode scans with the DBB were performed. Due to thermal lensing, the beam will be slightly astigmatic and so optimising the mode matching from the amplifier to the DBB was difficult. When optimising the pump-diode currents, the thermal lensing changes, so the mode matching to the DBB needs to be corrected for this.

4.5 Intensity Stabilisation of a 100 W Beam

The free running-intensity noise of the laser exceeds the level needed for the interferometer to reach its design sensitivity. Therefore, the laser must be stabilised. As actuators on the lasers intensity, a current shunt and an AOM were compared. Using the current shunt rather than the AOM enabled 10 W more power to be obtained in the beam emerging from the PMC, however the behaviour of the current shunt was highly dependent on the DC pump-diode current at 10 Hz where the laser noise requirement is most strict. If the amplifiers were required to operate under different

circumstances during the course of an observation run, the current shunt may prove to be highly unreliable. In contrast, the AOM has proved reliable during previous observation runs. While the AOM causes the beam to have less DC power, this may not be a problem for O4 as it is unlikely that the full 125 W will be used due to point defects on the test masses.

4.5.1 The Concept and Requirements for Intensity Stabilisation Servo

The intensity noise requirement for LIGO's laser is found in [1] and reproduced in Figure 4.11. At low frequencies, the requirement is driven by the need for the intensity noise to be less than the radiation pressure noise induced by the difference between the arm powers. At 10 Hz, the laser must be stabilised to a level of $\sim 1 \times 10^{-9} \frac{1}{\sqrt{\text{Hz}}}$. Above 10 Hz, the requirement on the intensity noise is relaxed as there is passive filtering of the light by the PRC. The light is passively stabilised by the PMC to reduce the intensity noise of the laser at RF frequencies.

The active intensity stabilisation of the laser is done in two stages; this is shown in Figure 4.12. First, the laser is stabilised by the inner loop to a level required to lock the IMC. The light used to measure the intensity noise for the inner loop is a low power beam that is transmitted by the PMC. There is ~ 1 mW of light in this beam, so the laser cannot be stabilised beyond the shot noise associated with a coherent state. For ~ 1 mW this is $\sim \frac{2 \times 10^{-8}}{\sqrt{\text{Hz}}}$. Once the IMC is locked, a pick-off from the beam entering the interferometer is used to measure the intensity noise of the laser. This beam has enough power, ~ 350 mW, to reach the level required, $\frac{1 \times 10^{-9}}{\sqrt{\text{Hz}}}$, during observation mode. The noise of the beam entering the interferometer must

be measured just after the IMC as this contains the intensity noise which cannot be measured by the Inner-loop diode e.g. the noise due to the periscope immediately after the PMC.

4.5.2 Free Running Noise of the Prototype Laser

To determine the open-loop gain needed for the servo to suppress the noise to the requirement, the free-running noise of the laser must be measured. The noise at different points along the beam path was measured to see the contribution of the NPRO, each amplifier and the PMC to the total noise.

The power noise of the NPRO was measured using the DBB RIN photodiode (Figure 4.13). The NPRO had a typical noise level for this type of laser [20]. This was recorded via CDS, so the measurements near 10 kHz are affected by the anti-aliasing filter. The noise between 100Hz and 1kHz had features associated with the motion of the table and mirror mounts.

The power noise of the first and second amplifiers was measured using the DBB RIN photodiode (Figure 4.13). The noise was measured with a spectrum analyser rather than CDS so that measurements up to 100 kHz could be made. As the amplifiers were pumped with low noise diodes and were operating in their saturated regimes, the RIN is expected to decrease after each amplifier. When an amplifier is saturated, the amplifier will not behave linearly and instead will add a fixed amount of input power regardless of its seed power. As expected, at frequencies above 10 kHz the RIN after the second amplifier was less than the RIN after the first amplifier. There was a peak at 20Hz; this can be attributed to the air conditioning.

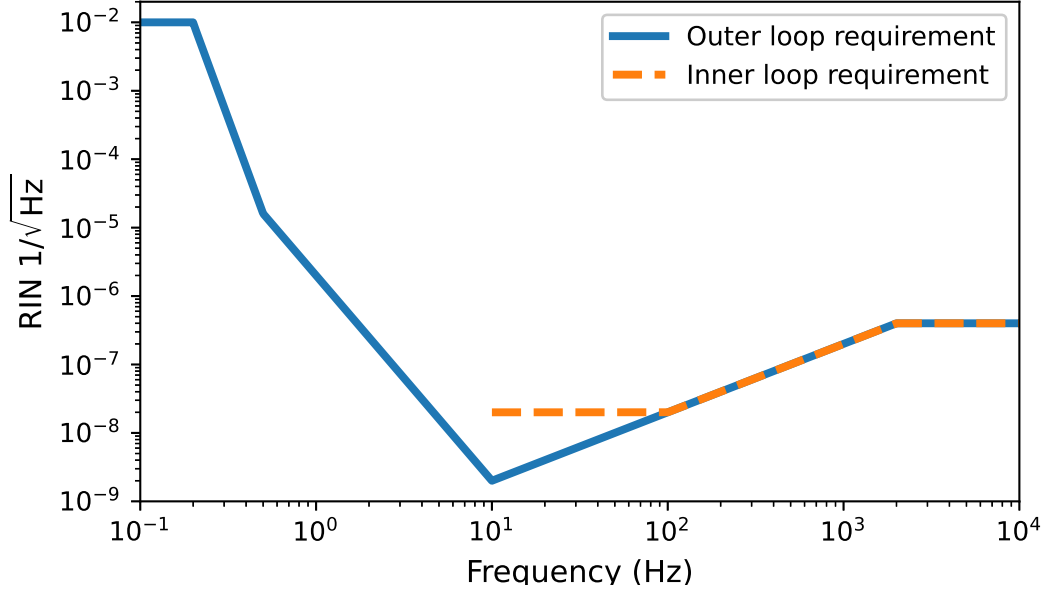


Figure 4.11: A reproduction of the intensity noise requirement from [1]. The outer loop requirement is shown in blue and the inner loop is shown by the dashed orange line. To lock the IMC, the inner loop requirement must be met. At frequencies below 10 Hz, the outer loop requirement is determined by the radiation pressure noise that is expected to arise due to imbalances in the arm power. The requirement is most stringent at 10 Hz. The requirement relaxes above 10 Hz due to the PRC cavity pole. The inner loop requirement is less extreme than the outer loop as it is only used to stabilise the laser in the initial locking sequence.

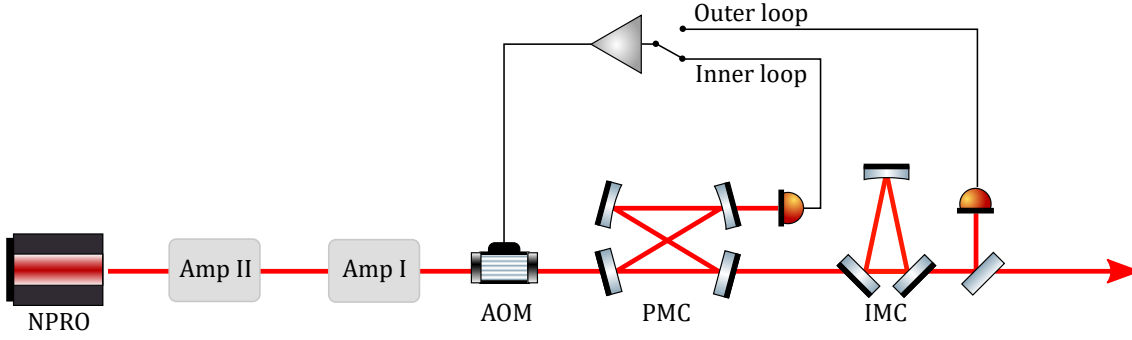


Figure 4.12: The intensity stabilisation concept involves two loops. The inner loop is responsible for stabilising the laser so that the IMC can lock. The outer loop is used to stabilise the laser for observation mode. The two loops sense the laser intensity noise at different points. The inner loop uses a low power beam, ~ 1 mW, transmitted by the PMC. The outer loop uses a higher power, ~ 350 mW, pick-off from the main beam. An AOM will be used to actuate on the laser's power.

The noise was covered in a forest of lines that were spaced 4Hz, starting from 4Hz and continuing up to high frequencies. These lines were found to originate in the Beckhoff slow-control computer which is used as an interface for the amplifiers. The laser control box needs better electrical isolation between its digital and analogue sections to remove the 4 Hz lines.

The noise of the beam emerging from the PMC was measured (Figure 4.13). The PMC introduces length to intensity noise coupling via the offset in the PDH loop. The offset needed to remove this coupling drifted due to changing levels of RF pick-up in the PMC servo electronics. The coupling from length noise of the PMC to intensity noise was decreased as much as possible.

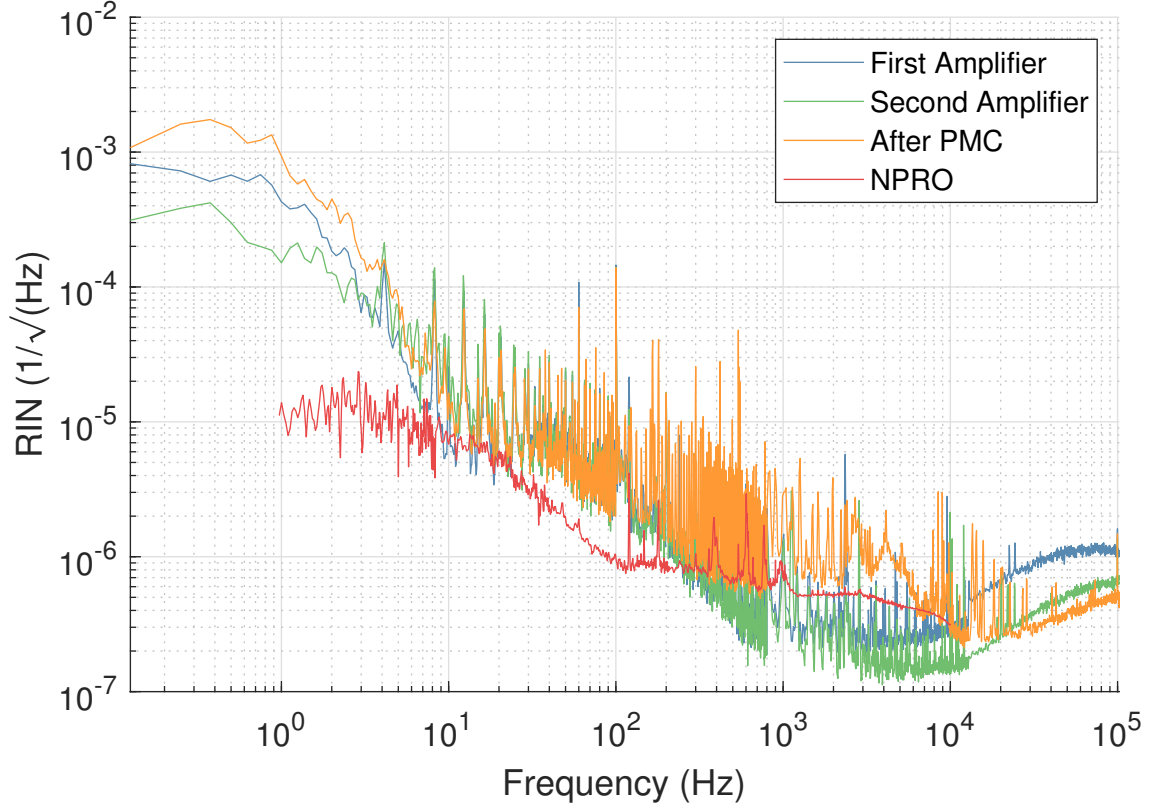


Figure 4.13: The free running noise of the NPRO and each amplifier is shown. The noise after the PMC was also measured. There is a forest of 4 Hz lines due to the Beckhoff control computer in the amplifier and PMC noise curves. Due to the non linear nature of the saturated amplifiers, at high frequencies the output beam intensity noise decreases relative to the input beam. The PMC couples length noise to intensity noise between 100 Hz and 10 kHz. This excess can be tuned away in the PDH servo.

4.5.3 Power Control Actuators

The actuator that will be used to provide intensity stabilisation will be an AOM. The use of a current shunt integrated into one of the pump-diodes in the second amplifier was investigated. The current shunt had the potential to reduce the number of transmissive optics in the beam path and increase the total power. Despite this, the current shunt may be an unsuitable choice for stabilising LIGO's laser. The current shunt transfer function depends on the DC current, at 10 Hz it changed by ~ 10 dB when the DC current was changed by ~ 1 A (see Figure 4.17), and so this may make the intensity stabilisation servo unreliable if the DC pump-diode currents needed changing over the course of an observation run.

4.5.3.1 Acousto-Optic Modulator

An AOM uses acoustic waves to scatter light into higher order modes. The amount of light that is Bragg scattered depends on the intensity of the acoustic wave. The AOM used was a Crystal Technology 3080-199. An RF driver, gooch and housego r21080-2am-rfx, was used to control the intensity of the acoustic wave. This driver produced an 80 MHz signal whose power can be modulated. The AOM's response was linearised with an IC (AD734). By modulating the power of this signal, the power in the zeroth order beam can be controlled. The diffraction efficiency of this AOM was found to be 74%. This means that the AOM can be used for coarse and fine power adjustment. To modulate the intensity of a laser beam, some of it must always be scattered into the first order beam. To operate linearly, the DC level of the light that is dumped was ~ 10 W.

The transfer function, in relative intensity units, between the AOM control electronics to the intensity of the zeroth order beam as measured by the ISS photodiode is shown in Figure 4.15. The schematic for this measurement is shown in Figure 4.14. This transfer function is flat from 1 Hz to 100 kHz, and, as the servo's unity gain point will be between 10-100 kHz, it is worth noting that above 10 kHz there is some phase lag.

4.5.3.2 Current Shunt

Modulating the current through a laser diode will modulate the intensity of the laser. This is a standard way of implementing intensity control at the 10W level [24–28]. Using a current shunt as an intensity actuator has been demonstrated in a similar system at the 100W level [13]. The current shunt technique allows for one to have fewer transmissive high power optics in the beam path, which potentially can reduce noise and thermal distortions of the beam. It requires a smaller DC offset compared to an AOM in order for it to work, therefore higher beam power can be achieved.

The amplifier's pump-diode current determines the amount of power produced by the amplifier. By modulating the current through one of the pump-diodes, the power of the laser can be stabilised. The current shunt consists of a power MOSFET and a resistor. It diverts current from the pump-diode by pulling it through the resistor instead. The amount of current that is pulled through the resistor depends on the voltage at the gate of the MOSFET (see [29, Chapters 2 and 3]).

The transfer function between the current shunt input and the ISS photodiode in transmission of the PMC was measured for a variety of different DC pump-diode

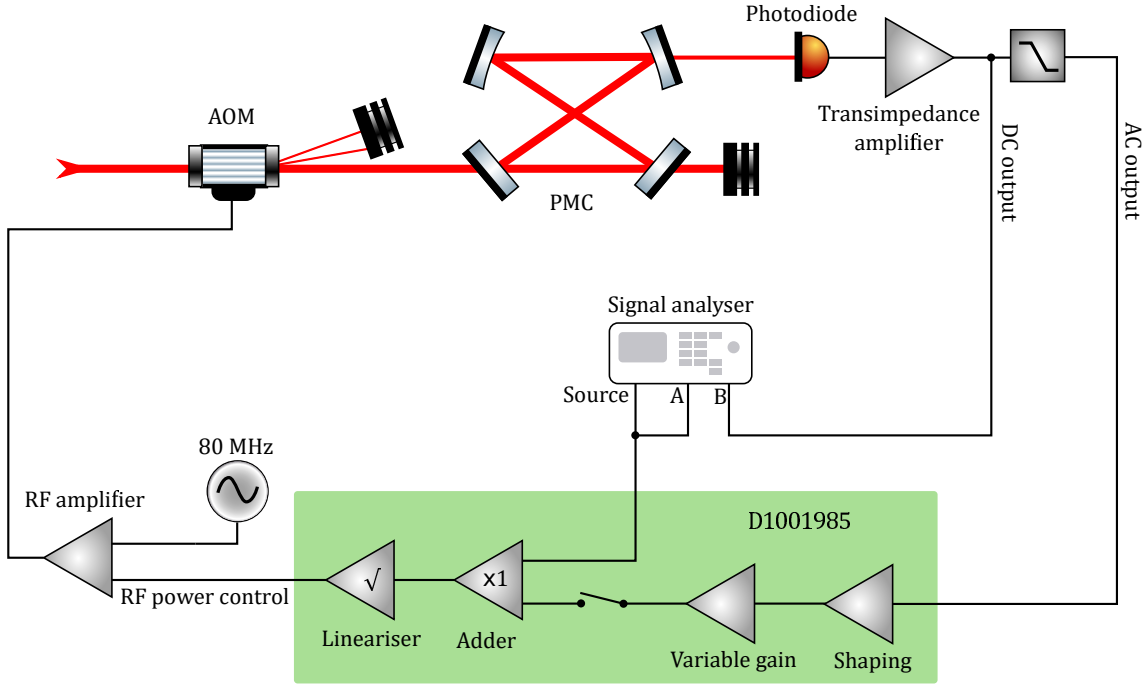


Figure 4.14: A sketch of how the AOM transfer function was measured. The components of the intensity stabilisation servo has also been shown. The PMC was locked and a signal analyser was used to perform a swept sine measurement. An IC (AD734) that performs a square root operation to an input signal was used to linearise the AOM's effect on beam power to an input signal. The transfer function was measured between the input to the lineariser and the photodiode used for the intensity stabilisation. The servo electronics are detailed in D1001985 and D1001998.

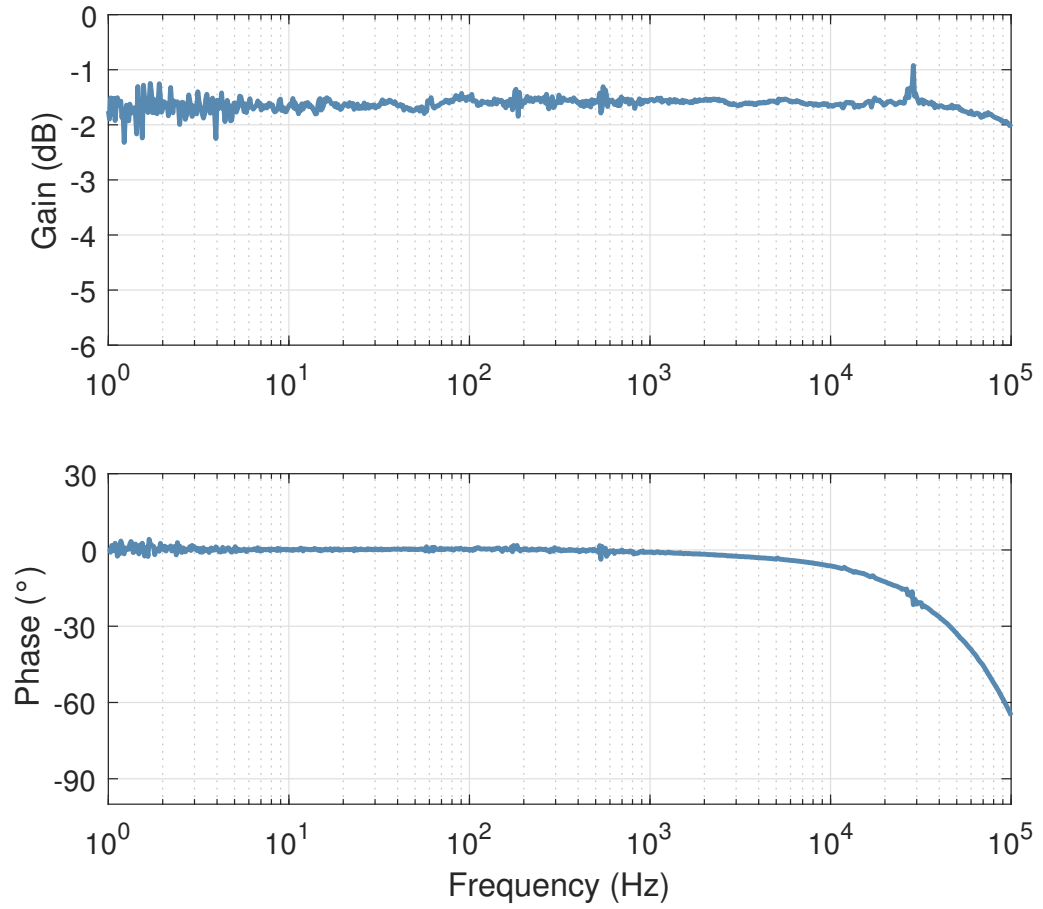


Figure 4.15: Using the layout shown in Figure 4.14, the transfer function between the input to the linearisation stage before the AOM to the light power change on the ISS photodiode was measured. The response was normalised by dividing by the DC power on the photodiode. The transfer function is flat over the region where the open loop gain needs to be greater than one (~ 10 kHz). There is some phase lag between 10-100 kHz; this is where the unity gain point of the servo will be and so this needs to be accounted for when determining if the loop will be stable.

currents. This is shown, in relative intensity units, in Figure 4.17. A schematic of the measurement is shown in Figure 4.16. The transfer function was found to have significant amounts of variation in gain and phase between 1 Hz-100 kHz. Above 10 kHz the transfer function did not vary much with DC pump-diode current. This is good, as the unity gain point of the intensity stabilisation servo will be between 10-100kHz. The AOM is a much stronger actuator than the current shunt; this can be compensated for by changing the servo electronics. It was found that the behaviour of the actuator at 10 Hz, where the servo requirement is hardest to meet, strongly depended on the DC pump-diode current.

4.5.3.3 The Reason for Choosing the AOM Instead of the Current Shunt

While the current shunt offers some advantages over the AOM, it cannot reduce the power at the many tens of Watts level. This is a functionality that LIGO would likely need as the detectors have never operated at full power, at present the LIGO detectors operate with between 25-50W, due to point absorbers on the test masses. The pump-diode DC current would have to be reduced to reduce the power by many tens of Watts. This would affect the noise of the laser and the open loop transfer function of the servo. These amplifiers work best in their saturated regimes and the RIN may be increased if the pump-diode currents are decreased significantly. There may be other circumstances in which the pump-diode currents need altering; towards the end of their life-time the diodes need to be frequently adjusted. Alternatively, high power attenuation optics would be required. Unlike the current shunt, the AOM's response depends on the DC light level in a trivial way. The AOM causes a decrease in total power, but this downside is outweighed by the reliability of the AOM compared to the current shunt. When the LIGO detectors are able to operate

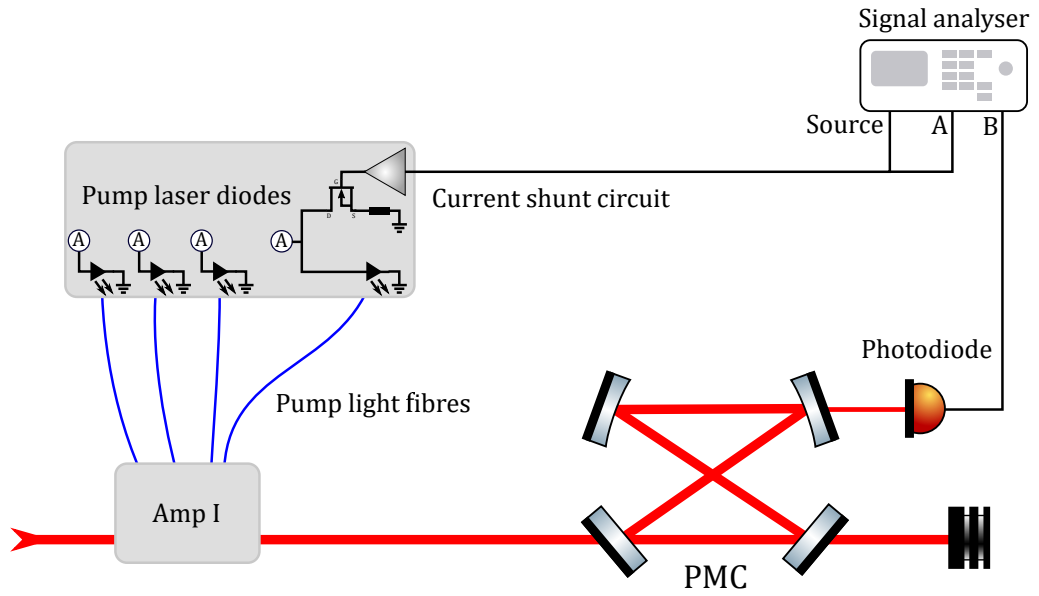


Figure 4.16: A sketch of how the current shunt transfer function was measured. The PMC was locked and a signal analyser was used to perform a swept sine measurement. The current of one of the pump-diodes in the second amplifier was modulated by applying a signal to a current shunt. The transfer function was measured between the input to the current shunt electronics and the photodiode used for the intensity stabilisation.

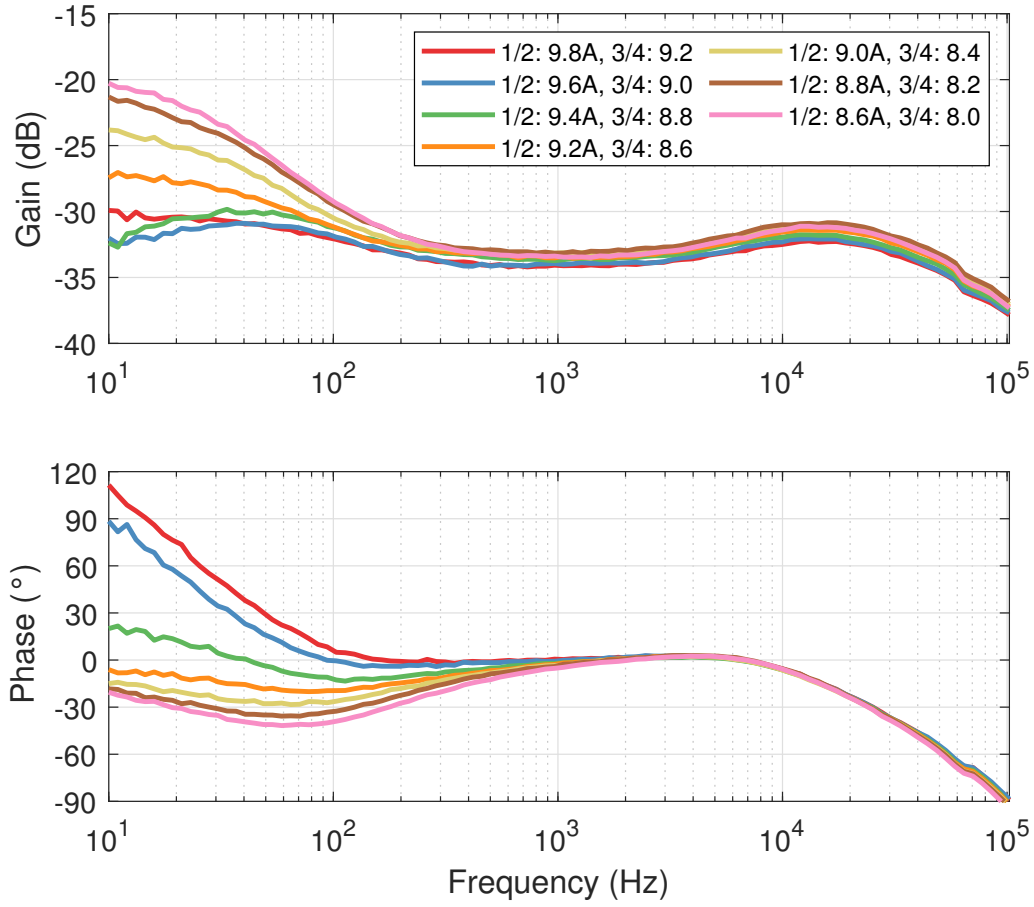


Figure 4.17: Using the layout shown in Figure 4.16, the transfer function between the current shunt input electronics and the intensity stabilisation photodiode was measured. The transfer function was normalised by dividing by the DC power on the photodiode. At 10 Hz, where the noise requirement is hardest to meet, the gain of the transfer function changes significantly depending on the pump-diode's DC current. The phase at this frequency does not matter as it is not near the unity gain point of the servo.

at full power, then the current shunt actuator would be worth revisiting

4.5.4 Open Loop Transfer Function of the Intensity Stabilisation Servo

The free running noise of the laser transmitted by the PMC is shown in Figure 4.13 and the requirement is shown in Figure 4.11. Based on the noise and the requirement, the unity gain point of the servo must be above 10 kHz. At 10 Hz, where the requirement is hardest to meet, the noise requirement is $2 \times 10^{-8} \frac{1}{\sqrt{\text{Hz}}}$ and the noise is $1 \times 10^{-5} - 1 \times 10^{-4} \frac{1}{\sqrt{\text{Hz}}}$, so there needs to be at least 60 dB of open loop gain at this frequency. If the noise of the laser was not dominated by the 4 Hz lines, the amplitude stabilisation could be achieved with a single pole low pass filter. However, this servo uses additional shaping to ensure that there is enough gain to suppress these peaks.

A sketch of the servo is shown in Figure 4.18. The servo features two circuits that add shaping; these are found built into the photodiode circuit [30] and within the rack electronics [31]. The servo model is described in detail in Section 4.5.4.1. The inner loop ISS features two photodiodes, one for in-loop measurements and one for out-of-loop measurements. The rack electronics allow for the servo to be controlled via CDS. This gives the user the ability to turn the servo on/off, to vary the overall gain and activate extra integrators.

A swept sine was applied at a times one adding stage of the servo to measure the open loop transfer function. The signal at the other input to this stage and the signal immediately after this stage was measured. The ratio of these two gives the

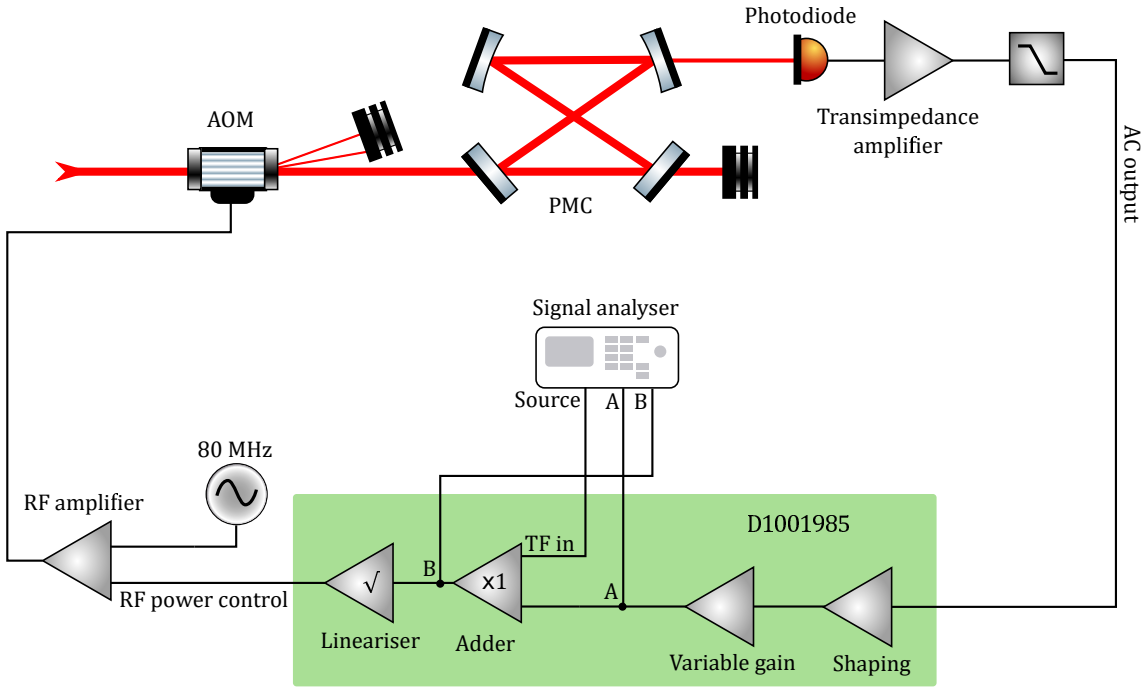


Figure 4.18: Sketch of the amplitude stabilisation servo. The open loop transfer function was measured by injecting a signal into a $\times 1$ adding stage and measuring the ratio of the signal immediately after and just before this stage.

open loop transfer function of the servo. For this measurement, the gain in CDS was set to 20 dB. Once the intensity stabilisation was activated, the integrators were engaged to increase the open loop gain at low frequencies. The light transmitted through the PMC was 100 W. The open loop transfer function is shown in Figure 4.19. The unity gain point is at 48 kHz with $\sim 70^\circ$ of phase margin. This servo satisfies the requirements.

4.5.4.1 Model of the Open-Loop Transfer Function

To understand the behaviour of the servo at frequencies where the measurement of the open loop gain saturated, a model is required. The overall gain of the servo is de-

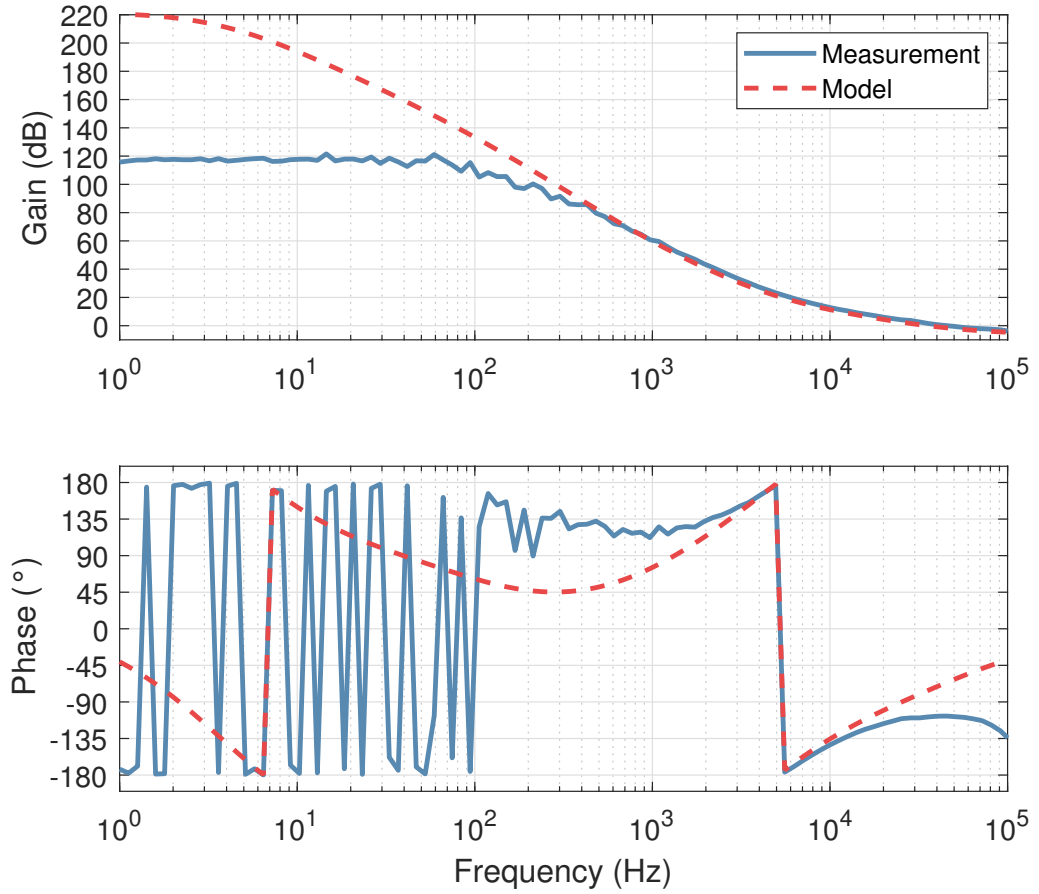


Figure 4.19: The open loop transfer function was measured using the layout shown in Figure 4.18. The unity gain point is at 48 kHz with 70° of phase margin. The shaping of the servo is $1/f$ at the unity gain point, but to achieve higher gain at low frequency, the servo has a $1/f^3$ slope below the unity gain point. The measurement saturated below 100 Hz. The model and measurement do not agree well below a few kHz due to the measurement saturating.

terminated by how much light there is on the in-loop photodiode and the setting of the variable gain. The following description puts all the gain that is distributed over all the components of the servo into one block. The photodiode transimpedance amplifier has a low-pass corner frequency of 5.9 Hz. This passes through a filter which boosts the photodiode's response to AC signals; this filter consists of two transitional differentiators with corner frequencies at 0.05 Hz and 3 Hz. The rack electronics feature two transitional integrators and one transitional differentiator. The first transitional integrator has corner frequencies at 1.06 kHz and 4.83 kHz. The second transitional integrator has corner frequencies at 1.06 kHz and 4.42 kHz. The transitional differentiator has corner frequencies at 59 kHz and 2.9 MHz. Activating the extra integrator stages removes the zeros in the transitional integrators at 1.06 kHz. The AOM introduces phase lag via a delay between the input and the output of 2.5 μ s. This model is summed up in Table 4.1 and shown in Figure 4.19.

4.5.5 Measurement of the Intensity Stabilisation of a 100 W Beam to the Inner Loop Requirement

The PMC was set to transmit 100 W of intensity stabilised light. The stabilised laser noise is shown in Figure 4.20. A noise budget was created using the signal measured on the in-loop photodiode and the shot noise level of the light used to stabilise the laser. This agrees with the out of loop noise measurement above 10 Hz. Below 10 Hz, there is excess noise in the out of loop measurement compared to the noise budget. Possible sources of this noise could be scattered light or dust falling through the ISS beam. Near the unity gain point, due to the open loop transfer function phase and gain, there is a small increase in the stabilised noise compared

Component	Transfer function
Photodiode low pass	$\frac{36.945}{s+36.95}$
Photodiode filtered output	$\frac{3600(s+0.3142)^2}{(s+18.85)^2}$
Rack transitional integrator I	$\frac{0.21946(s+3.035e04)}{(s+6660)}$
Rack transitional integrator II	$\frac{0.023982(s+2.777e04)}{(s+666)}$
Rack transitional differentiator	$\frac{49.153(s+3.707e05)}{(s+1.822e07)}$
Rack additional integrators	$\frac{1 \times 10^{-8}(s+6660)^2}{(s+0.666)^2}$
AOM phase lag	$\exp(-2.5 \times 10^{-6} s)$
Overall gain	2.6775×10^{10}

Table 4.1: Model of the intensity stabilisation servo expressed in terms of the complex angular frequency $s = \omega + i\sigma$. Note that in reality the gain is distributed amongst each block; in this model all the components have unity gain at DC. The additional integrators have been expressed as a separate block to the transitional integrators with a pole that cancels a zero in the transitional integrators; the zero in this additional integrator transfer function was chosen to be below any frequency of interest and does not exist in the servo. It was included here to make modelling the servo more clear.

to the free running noise, however this is out of the requirement frequency band and so this is not an issue. This set-up will be able to deliver at least 100 W of intensity stabilised light, which will likely be more than is required at the time of writing, to the interferometer.

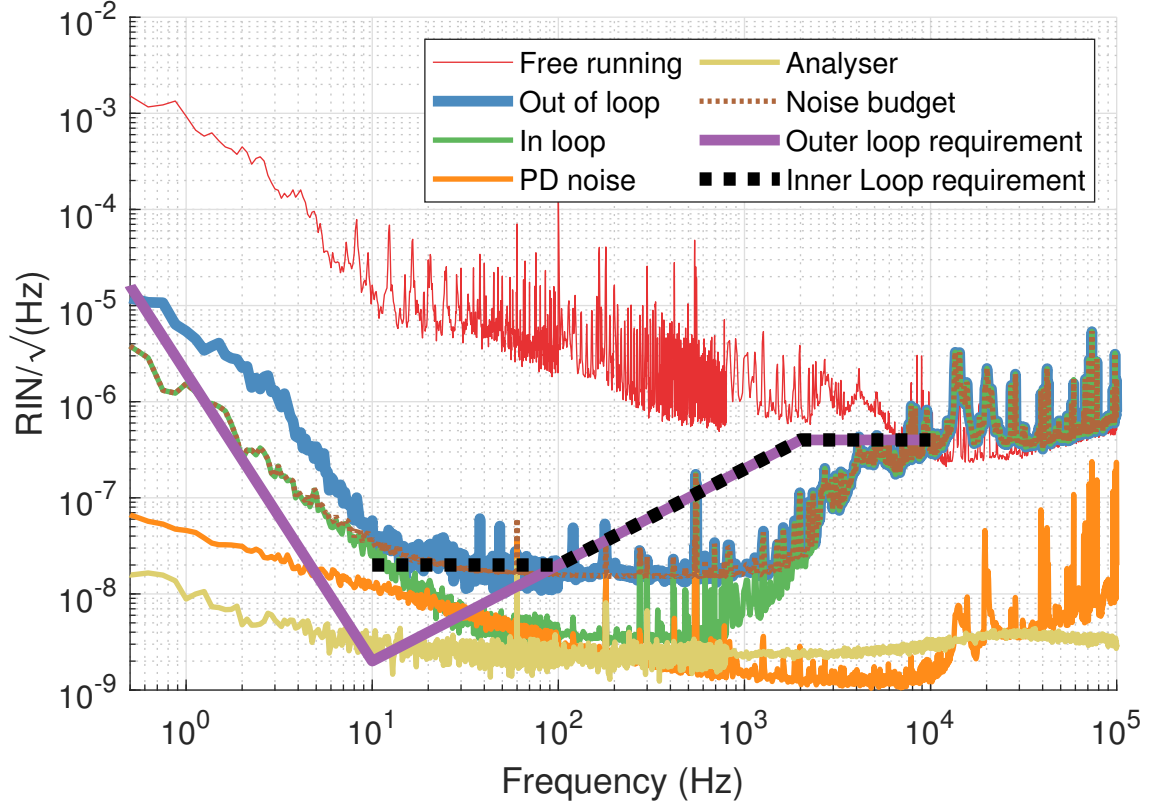


Figure 4.20: The free-running noise of the laser is shown in red. The noise of the light transmitted by the PMC is shown in blue; this meets the requirement that is shown by the dashed black line. The noise level is determined by the in loop noise, shown in green, the shot noise and other technical noise sources. The sum of these is shown by the dotted brown line. Below 100 Hz, the noise is shot noise limited. There is a discrepancy between the noise budget and the measured noise. This is likely due to dust falling through the beam or scattered light. Near the unity gain point of the servo, the noise is enhanced by the servo due to gain peaking.

4.6 Pointing Noise of the Beam Entering the PMC

The pointing noise witnessed at LHO is one of the reasons why single-pass amplifiers were chosen as the replacement for the HPO. Pointing noise manifests as intensity noise since a change in beam direction can be modelled as the introduction of higher order modes. These new amplifiers are cooled in such a way that the jitter on the PSL beam is reduced. Pointing is passively stabilised by the PSL via the PMC. The pointing noise requirement is based on the transfer function of the HG10 mode between the input of the interferometer and the dark port [32].

The pointing noise was measured using the DBB. The pointing noise is shown in Figure 4.21. To measure pointing noise, the laser was locked to the DBB cavity. The reflected light from the cavity was made up of the fundamental mode of the cavity (some of this mode leaks out) and the first order modes associated with misalignment. The beat between these two fields can be used to determine the pointing of the beam; this is known as differential wavefront sensing [33]. The electronic noise floor is at $1 \times 10^{-7} \frac{1}{\sqrt{\text{Hz}}}$. The pointing is expressed in units of $p = \delta x/w_0 + i\delta\alpha/\Theta$, where Θ is half the divergence angle of the beam, α is the angle of the beam, x is the beams position and w_0 is the beams waist.

The pointing noise measurement shows that this layout is close to fulfilling the requirement. More time could be used to reduce turbulence in the water cooling as this can introduce jitter. The noise below 100 Hz may be coming from the environmental control in the TnT lab. The LIGO PSL table is different, so this data may not reflect how quiet this layout will be in the main lab.

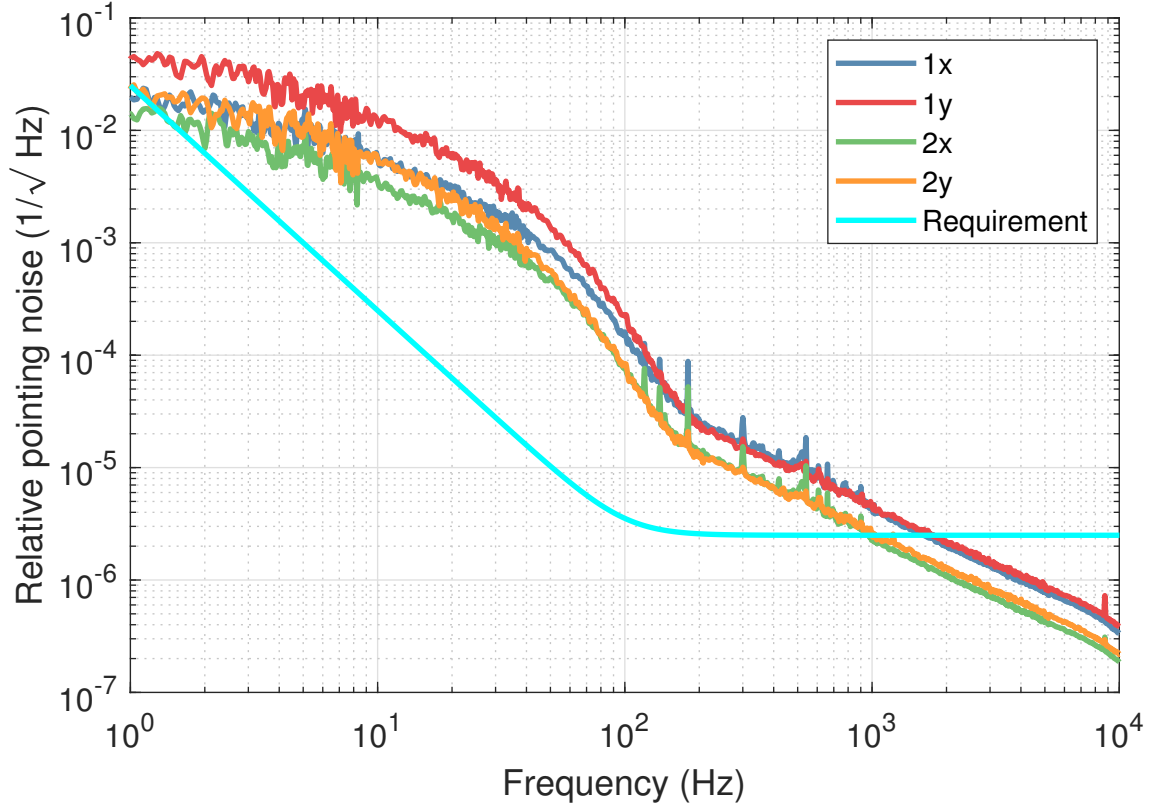


Figure 4.21: The four relative pointing noise degrees of freedom for the beam into the PMC was measured with the DBB. The requirement [32] is shown in light blue. The pointing noise is above the requirement, however the TnT lab has a higher amount of environmental noise; this pointing noise will be less in the PSL. Measurements below 100 Hz may be affected by scattering.

4.7 Free Running Frequency Noise of the Prototype Laser System

The LIGO detectors employ a multi-stage frequency stabilisation system. Ultimately, the arm cavity is used as a frequency reference since $\delta f/f = \delta l/l$, however the initial stabilisation of the laser uses an ultra low expansion fused silica reference cavity located on the laser table [34]. The free-running frequency noise must be measured to design the frequency stabilisation servo.

The PMC can be used to measure the frequency noise of the beam emerging from the PMC. The beam's frequency will be determined by the length noise of the PMC and the frequency noise of the NPRO. The signals used to control the PMC can be converted into a measurement of the systems frequency noise. The control signal which is sent to the PZT in the PMC can give the frequency noise of the system where there is sufficient open loop gain. Around the unity gain point, the servo's response needs to be factored in. Alternatively, the error signal at the mixer and the open loop gain can be used to measure the frequency noise of the system.

The NPRO is locked to the PMC by controlling the length of the PMC. The length of the PMC is controlled with two actuators. A PZT is attached to one of its mirrors to provide fast control, and a heater which determines the thermal expansion of the PMC provides slow control. The PDH technique is used to sense the length of the cavity. RF frequency side-bands are applied to the laser via an EOM. The light that is reflected from the PMC is demodulated at the same frequency as the RF sidebands. When the cavity is near resonance the demodulated signal is proportional

to the difference between the laser frequency and the frequency corresponding to the cavity length. A sketch of the PMC length control servo is shown in Figure 4.22.

To see how the error signal used to lock the cavity and the open-loop transfer function gives the frequency noise, consider Figure 4.23. The difference between the laser frequency and the cavity frequency is obtained via the PDH technique. The PZT signal, c , is then obtained via the transfer function of the optics, i.e. the frequency to voltage transfer function corresponding to the slope of the demodulated PDH signal, and the transfer function of the electronics. Together this transfer function is G . First, define the error signal $c = a - b$. Next, express the control signal in terms of the error signal, $cG = d$. Expressing the signal b in terms of d allows for signal a to be written in terms of the error signal c ,

$$a = c(1 + GH). \quad (4.1)$$

Equation (4.1) shows that to measure the frequency noise of the laser, a , the error signal should be, approximately, multiplied by the open loop transfer function, GH .

4.7.1 Calibrating the Mixer Signal, the PZT and Modelling the Open Loop Transfer Function for Locking the PMC

As the cavity is short compared to the frequencies of interest, the transfer function representing that converts a difference in NPRO and PMC frequency, the signal due to the PDH scheme, to a voltage is a flat gain factor. A measurement of this can be obtained by measuring the slope of the PDH signal as the frequency of the cavity

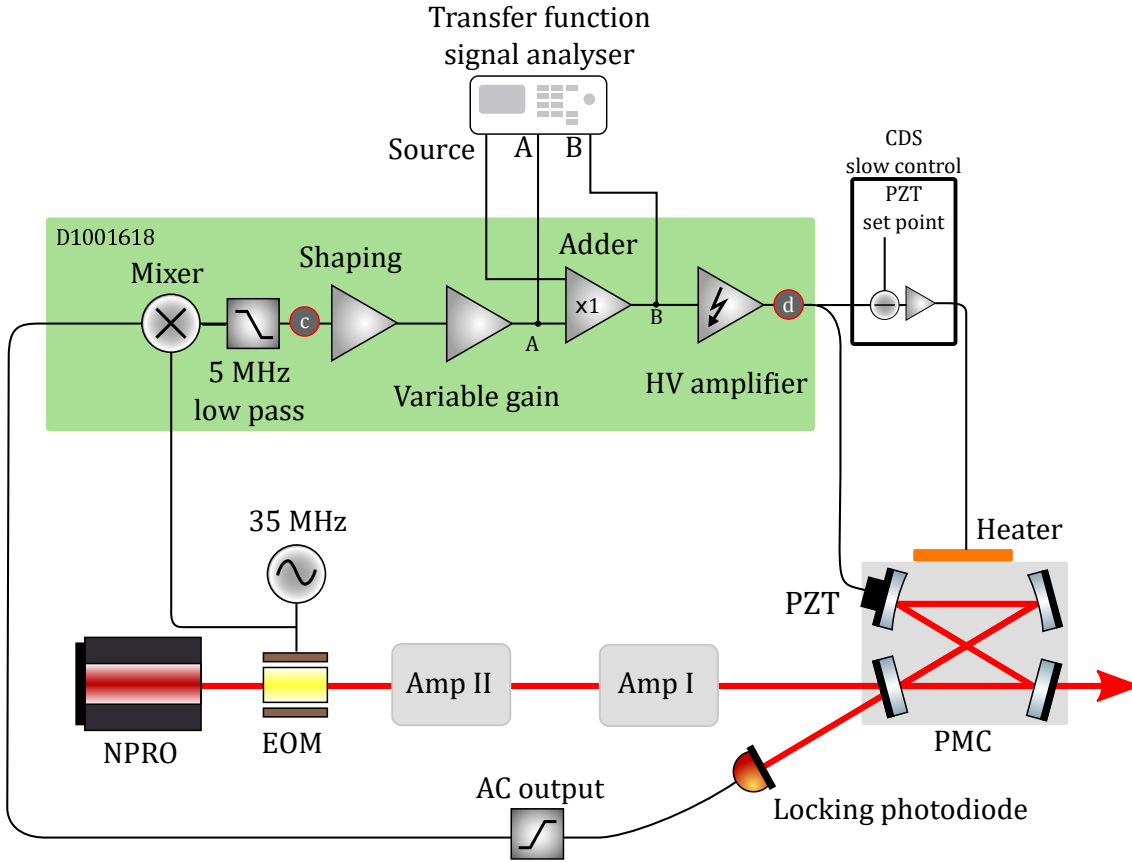


Figure 4.22: A sketch of the frequency stabilisation servo. An EOM creates 35 MHz sidebands which are used to produce a PDH locking signal. A PZT controls the length of the cavity at high frequencies. A heater and a digital loop is used to provide slow control that corrects for frequency drifts associated with change in the NPRO temperature and the room temperature. This is required as otherwise the PZT would run out of range. The open loop transfer function was measured using signals at A and B. The frequency noise was measured using the error point and control signals marked by *c* and *d*, with this notation corresponding with the notation in Figure 4.23 and Equation (4.1).

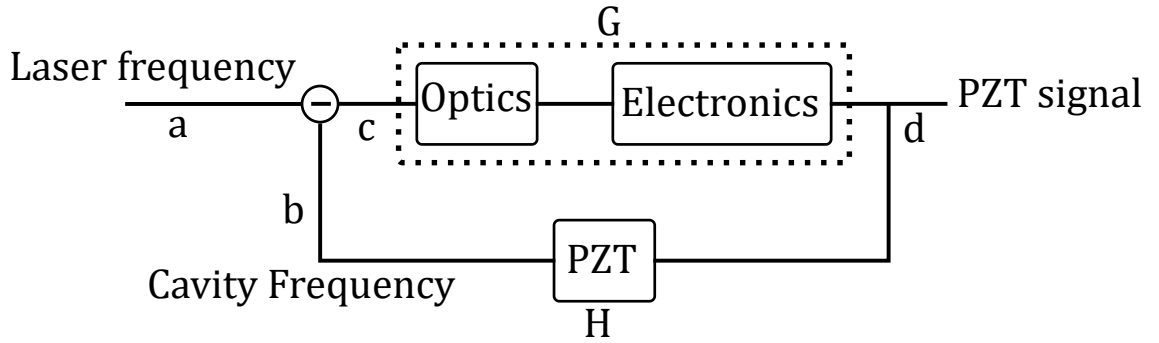


Figure 4.23: The servo can be considered as being made of three parts: the optics, the shaping electronics and the PZT. The transfer function of the optics represents how a difference in frequency between the laser and the cavity transforms into a signal after the demodulator. When at resonance and below the cavity pole, this is a flat transfer function that is proportional to the difference in frequency. Since this cavity pole is in the MHz region and this measurement will only go to the kHz region, the cavity pole can be ignored. The transfer function of the electronics includes everything after the demodulator up to and including the final high power OP amp. The PZT transfer function converts a voltage to a cavity frequency. The PZT acts as a low pass filter due to it having capacitance.

passes through a resonance. By driving the PZT with a ramp and simultaneously measuring the signal before and after it is demodulated, the transfer function of the optics can be obtained. Figure 4.24 shows this measurement. The time axis can be converted to frequency as the full-width-half-maximum (FWHM) of the reflected light is determined by the optical properties of the PMC. The design value of the FWHM for the PMC is 1.19 MHz. The slope of the demodulated signal as the cavity passes through the resonance can be used to give the conversion between difference in frequency of the cavity and laser to voltage after the demodulation electronics. This factor is 0.1364 MHz/V.

A ramp was applied to the PZT and the spacing between two resonances was measured to convert the PZT voltage to the cavity frequency at low frequencies. The difference in ramp voltage between the two resonances corresponds to the FSR of the cavity. This is shown in figure 4.25. The nominal FSR for the PMC is 148.529 MHz, so the PZT to frequency transfer function is 42.2870 MHz/V. The PZT forms a low pass filter as it has capacitance; this must be included when modelling the servo. The corner frequency of the PZT was estimated to be at 1700 Hz. The corner frequency of the PZT is above the unity gain frequency of the PMC locking servo.

The measurement and model of the open loop transfer function, i.e. the transfer functions of the optics, electronics and the PZT multiplied together, is shown in Figure 4.26. The model is within 2 dB of the measurement. A sketch of how the open loop transfer function was measured is shown in Figure 4.22. The unity gain point of the servo was measured to be at approximately 800 Hz. The difference between the model and the measurement may be due to the use of nominal values

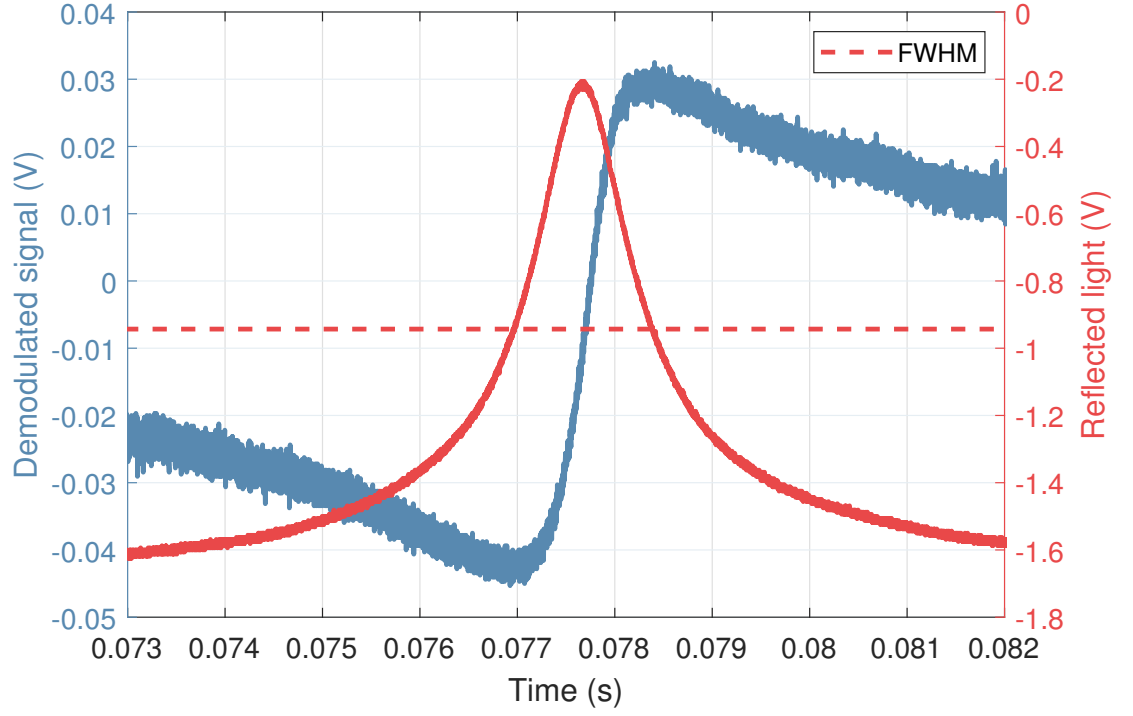


Figure 4.24: A ramp was applied to the PZT and the photodiode signal was measured before and after demodulation. The demodulated photodiode signal is shown in blue. The photodiode signal is shown in red. Around the peak of the photodiode signal, the slope of the demodulated signal gives the gain of the ‘optics’ transfer function. This can be expressed in units of MHz/V by converting the time axis to a frequency axis. This is done by using the FWHM of the photodiode signal as a reference frequency. The level of the FWHM is indicated by the dashed red line.

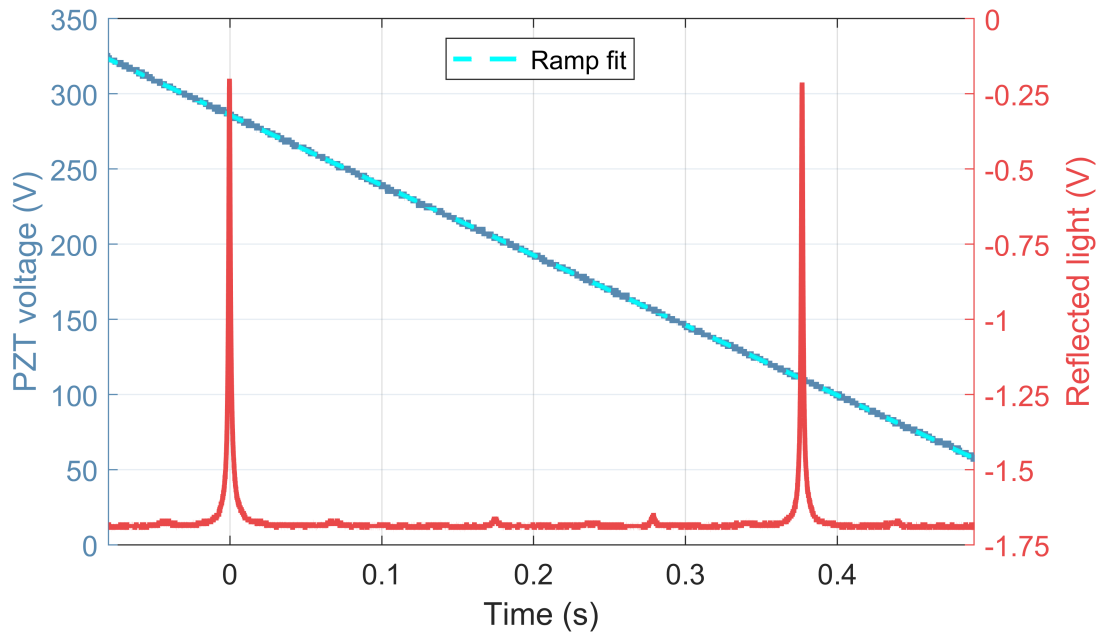


Figure 4.25: A ramp was applied to the PZT and the reflected light from the cavity was measured. The FSR of the cavity is known, so the time in this measurement can be converted to frequency. This gives the PZT transfer function at low frequencies. The ramp signal is shown in blue and the reflected light is shown in red.

for the properties of the PMC.

4.7.2 Frequency Noise Measurement

The measurement of frequency noise is shown in Figure 4.27. Despite imperfect calibration, this measurement provides useful information about the frequency noise of the laser. As the PMC was not vacuum sealed, between 100 Hz to 1 kHz, there was a bulge in the frequency noise. This was due to the air conditioning generating PMC length noise combined with non perfect offsetting in the PDH loop. Below 100 Hz and above 1 kHz, the frequency noise of the laser was determined by the NPRO. Near the UGP, there was gain peaking which will enhance the noise.

Below the unity gain point, as the PZT response is flat, the PZT signal can be converted into frequency noise. The mixer signal and the open loop transfer function $c(1 + GH)$ gives the frequency noise. These two measurements are within a factor of 2 of each other. This noise is higher than the expected level for the NPRO. There is uncertainty in the calibration as nominal values for the optical properties of the PMC were used.

At frequencies much above the unity gain point of the servo, a signal applied to the PZT will not be suppressed by the loop. Near the unity gain point, a signal applied to the PZT will be attenuated. A peak at 1111 Hz was used to check the calibration of the loop. The closed loop gain at 1111 Hz was modelled to be -5.2 dB, so the peak measured at the mixer should be reduced by a factor of $1 - 10^{-5.2/20} = 0.45$. The peak generated at the PZT was measured at the mixer to be reduced by a factor of 0.70; the difference between the prediction and measurement would be fixed with

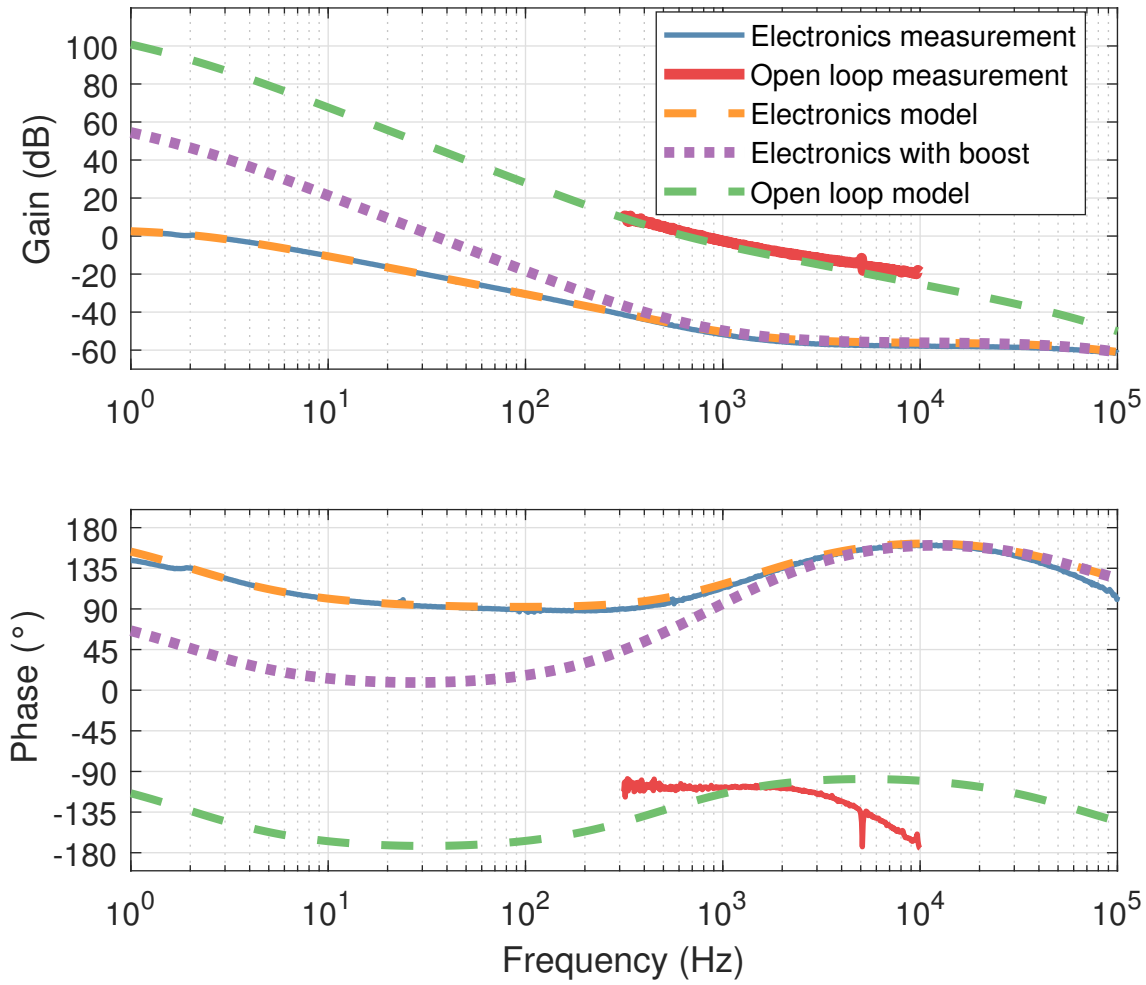


Figure 4.26: The open loop transfer function was measured using the layout shown in Figure 4.22. This is shown in red. The transfer function of the electronics without the ‘boost’ provided by an extra integrator stage was measured and is shown in blue. The model of this is shown in orange. The boosted electronics model is shown in purple. The green line shows the expected servo gain based on the boosted electronics model and the measurements shown in Figures 4.24 and 4.25. This model is within 2 dB of the measurement. The unity gain point was measured to be at ~ 800 Hz.

better characterisation of the PMC.

The discrepancy between the blue and red traces in Figure 4.27 may be due to the optical values used for the PMC, as these were not measured. If the cavity is detuned, a frequency to amplitude conversion occurs. This conversion does not happen at the FSR of the cavity. Measuring the transfer function between a frequency modulation generated by a broadband EOM and the amplitude of the light transmitted by the cavity will give an accurate measurement of the length of the cavity [35]. To measure the finesse of the cavity, an RF amplitude modulator could be used. The transfer function between the input modulation and the transmitted light depends on the FSR and the finesse of the cavity [35]. The AOM in the layout cannot be used as its bandwidth is too low. In Figure 4.27, the red trace below the unity gain point is better calibrated since the FSR does not depend on the cleanliness of mirrors.

4.8 Conclusion

The PSL has undergone a major redesign due to the jitter noise introduced by the cooling for the HPO. The new layout consists of an NPRO and two solid state amplifiers followed by the PMC. A prototype of this new design was constructed and tested. The set-up procedure was established and a cooling scheme was trialled.

The frequency noise of the beam transmitted by the PMC was investigated using signals in the PMC control loop. The laser is dominated by NPRO noise at frequencies below $\sim 100\text{Hz}$ and above $\sim 1\text{kHz}$. As the PMC is not shielded from acoustic vibrations, the PMC introduces noise between these two frequencies.

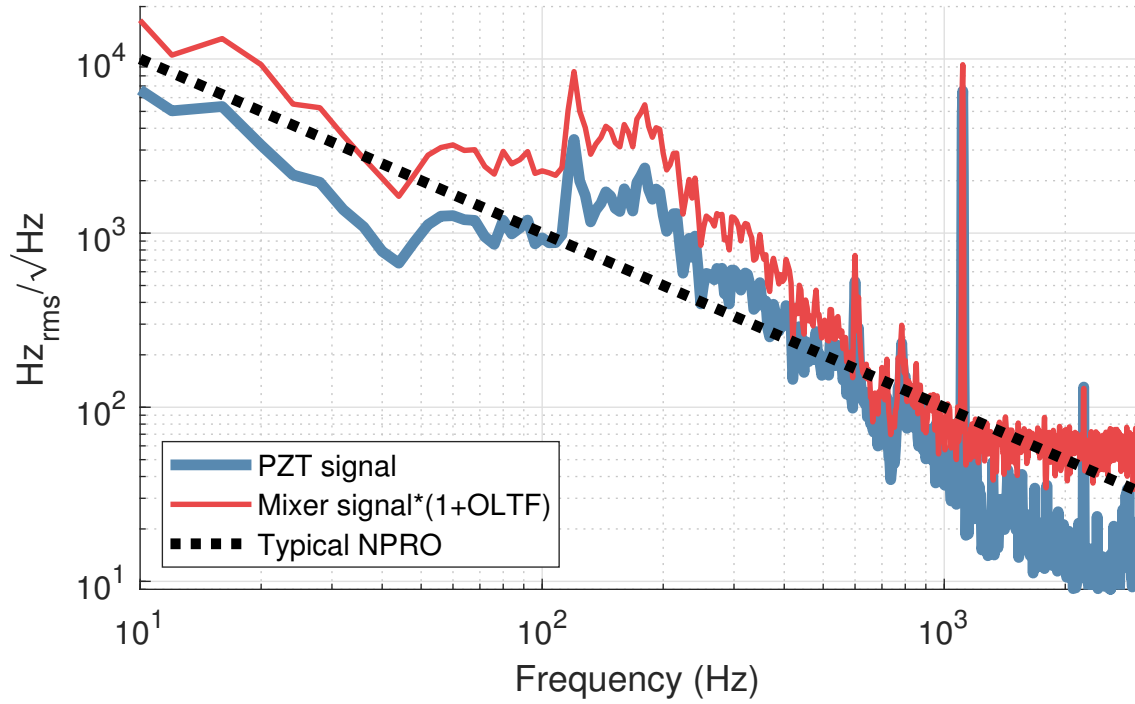


Figure 4.27: The mixer signal combined with the open loop transfer function is shown in red. This is equivalent to the frequency noise of the system. The PZT signal should agree with this measurement below the unity gain point (~ 800 Hz). A signal was applied to the PZT at 1111 Hz. This is below the corner frequency of the PZT and above the unity gain point for the servo. This should match with the mixer signal. There is a factor of two between these two measurements. This can be attributed to imperfect calibration due to nominal parameters of the PMC being used. The typical noise expected from an NPRO is shown by the dashed black line. Between 100 Hz and 1000 Hz, there is a bulge in the frequency noise. This is likely due to acoustically driven length noise of the PMC as the PMC is not vacuum sealed.

The jitter was found to be close to the requirements. As the PSL is in a quieter environment than the lab this layout was tested in, this may not be a problem. To reduce the jitter noise further, the flow rates being reduced may help, however this reduces the output power. Active alignment could be implemented using the light that is reflected from the PMC. Turbulence generated by bends in the water cooling pipes could be reduced to lower the jitter noise.

The output power was measured over two days and was found to drop from a peak of 111 W to 107 W. This was due to changes the lab temperature affecting the alignment. Higher output power could be achieved if more pump-diode current and cooling was used. Additionally, more power could be coupled to the PMC if more time was spent on optimising the mode matching between the second amplifier and the PMC.

100W of intensity stabilised light was transmitted through the PMC. The free running intensity noise of the beam transmitted by the PMC was measured and the inner loop intensity stabilisation was tested. Two actuators were considered, a current shunt and an AOM. The AOM was chosen as this offers the most flexibility in terms of loop gains and operating power. The inner loop noise requirement was met with this set-up.

Future iterations of the PSL may use fibre lasers and fibre amplifiers [10, 11] as there is rapid development in this technology. These have been demonstrated at the power levels required for LIGO. Coherent power addition is another technique that can be used to obtain high beam powers. When long term operation has been demonstrated, these technologies may be a viable replacement for the current laser

design.

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